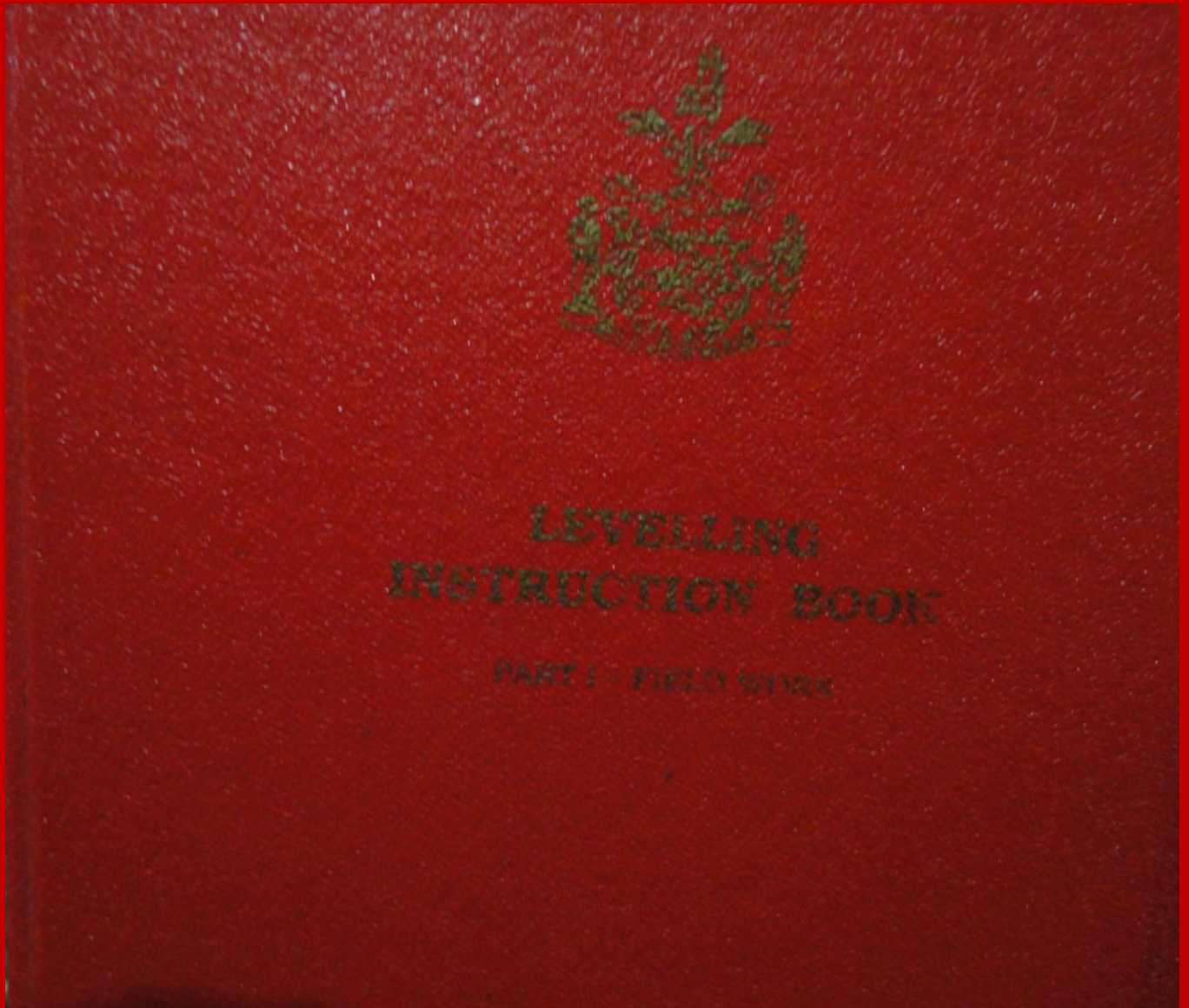




RED BOOK

KD Singh

2. TERMINOLOGY

Level Surface	is that surface which is at all points normal to the direction of gravity as indicated by a plumb line.
Levelling	is an operation for the determination of the relative difference of height between any two points on the earth's surface.
Level line	is a line which lies on one level surface and is therefore perpendicular to the direction of gravity at all points along the line.
Horizontal Plane	passing through a point is the plane normal to the vertical at that point and therefore is tangential to the level surface at that point.
Line (Ln)	is called the route followed by the method of levelling from point A to point B.
Loop (Lp)	is called the closed system of a certain number of lines.
Alignment (Al)	is called a sequence of coherent roads considered in one line. (e.g. The road Birganj - Pathlaiya - Hetauda - Naubise - Kathmandu is considered as one alignment).

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Datum Surface	is an adopted (assumed) level surface to which the elevation of points are referred. Mean Sea Level is most frequently used as Datum Surface.	levelling network. They are also established as special monumentations.
Mean Sea Level (MSL)	is the mean level of the sea at a selected place and is determined by taking the mean value of the height of the tide, measured at hourly intervals over a long period, with respect to a fixed mark.	
Height	of any point is its vertical distance over or below the reference datum surface.	
Bench Mark (BM)	is the common expression used for any kind of monumented reference mark of any material of which the height above the selected datum (Mean sea level) has been determined by the process of levelling.	Permanent Bench Mark (PBM) is called such a monumentation which outlasts at least the next 50 years without changing its relative position neither vertically nor horizontally. Such monumentations are established with a single BM or as a triple or quadruple group.
Datum Stations (DS)	are such BMs from which the height above Mean Sea Level for Nepal is derived. They are usually established as special-monumentations and are situated close to the boundaries of Nepal.	Junction Point (JP) is called the intersection of a line or the point where two lines cross together. At such points triple or quadruple groups of BMs are established.
Base Points (BP)	are such BMs which are established in geologically stable areas to serve always for the control and maintenance of the	Section is a line between any two permanent Bench Marks.
		Sub-Section is the distance within a line of such length that can be levelled with 6 to 8 instrument set-ups. i. e. sub-section is the sub-division of a section.
		Line of collimation (Line of sight) is the line joining the intersection of the cross hairs to the optical centre of the objective.
		Back sight is the first reading taken from any instrument position to a known height.

Fore sight	is the last reading taken from any instrument position.	height between two consecutive TBMs resulting from Forward and Backward Levelling. (Lev. Form 1.16)
Change point (Turning point)	is a staff position to which both a Foresight and a Backsight are taken.	
14-days section (14-ds)	consists of such numbers of sections that its length can be levelled within 14 days, by one team forward and at the same time, by the other team backward. Observations of 14-ds must always begin and end on a triple or quadruple group of PBMs.	is called such a BM, which can no more be found <i>after</i> its height was observed and computed. BMs which are discovered to be unstable or <i>lost at the time of the first</i> determination of their height, are not considered to be destroyed ones as they can be re-established and then treated as if nothing had happened.
Link (Lk)	is called such a line which is levelled for purposes other than for the establishment of a coherent levelling - network. Links depart and end always at PBMs. e. g. Heights of Trigonometrical Stations are derived from a PBM by means of a link.	Forward Levelling is the levelling carried out in the direction of the increment of the Bench Mark numbering (i. e. the forward direction.)
Temporary Bench Mark (TBM)	is called such a monumentation (usually with an iron rivet) which must outlast at least during the observation period. TBMs serve to check the accuracy of the sections as one compares immediately after finishing a 14-days section, the two differences of	Backward Levelling is the levelling carried out in the direction of the descending order of Bench Mark numbering. (i. e. in opposite direction).
		Destroyed BM

3. INTRODUCTION

Levelling is the process of the determination of the relative altitude of points on the earth's surface. The altitude or height of any point is its distance above the datum surface (Mean Sea Level i.e. Geoid). Various heights are in use, which are referred to their respective datum such as Spheroidal and Geoidal Height.

The principles of Precise or Geodetic Levelling are the same as ordinary spirit levelling. The only differences in the methods of levelling are the following conditions imposed on Precise Levelling to eliminate all the possible sources of errors.

(a) Instrument refinement:

The use of high precision instruments and more accurate invar staves.

(b) Field procedure refinement:

(i) The application of more rigorous observing techniques under restricted climatic and environmental conditions.

(ii) Refinement in booking and reduction of readings.

Thus, this process results in precise Geoidal Height if extreme care is taken during the process of observation and computation.

Applications of High Precision Levelling:

The H. P. Levelling is normally used for:-

- (a) Basic levelling network of a country.
- (b) Geodetic Research, Tectonic, Crustal Movement, Mean sea Level studies etc.

- (c) Frame work of vertical control for major engineering works, dams, bridges and canals etc.

Accuracy

Various survey agencies have their own specifications. The standard of accuracy required for Precise Levelling will vary according to the purpose for which the levelling is required. For the Survey Department/H. M. G., Nepal, to provide a field check, the following tolerances are specified as the maximum acceptable misclosure between Forward and Backward Levelling on any section of levelling.

ORDER	TOLERANCE	
High Precision/First	$1.1\sqrt{K}$ mm	where K is
Second	$2.0\sqrt{K}$ mm	distance in Km
Third	$5.0\sqrt{K}$ mm	

Network

In laying out primary levelling network in a country, the whole area to be levelled should be covered by the network. That network consists of traverses (lines) in the shape of circles, rectangles and triangles. The length of their sides will vary depending upon the nature of the terrain.

High Precision/First Order

When selecting lines of levelling for the H.P. Levelling, the following points should be borne in mind.

- (a) Lines should follow routes of communication but due consideration must be given to avoid the vibration caused by excessive traffic.

- (b) If at all possible, a large change of height should be avoided.
- (c) If it is feasible, the gap crossing (i.e. rivers etc.) should be avoided.

In running the H.P./First Order Levelling, the following requirements are fulfilled.

- (a) A High Precision/Geodetic Levels with parallel plate micrometers are used.
- (b) A pair of invar staves with circular bubbles are used.
- (c) Maximum line of sight = 25 m and
Backsight Distance = Foresight Distance
- (d) Staff reading below 0.5m and above 2.5m are not to be used.
- (e) Observation sequence should be followed strictly.
- (f) Forward and Backward Levelling are to be carried out by different equipment and observers.
- (g) All the possible sources of errors must be minimised and subsequent corrections are to be applied.

First Order Levelling network provides the basis for all other height control in a country.

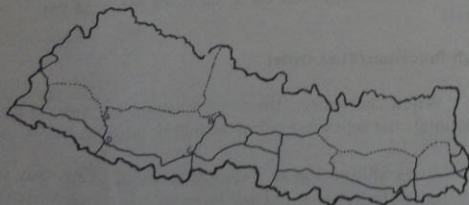


Fig. 1

- Second Order:**
- i.e. sub-division of First Order
 - Less geodetic corrections than for the First Order Levelling.
 - Shorter lines than First Order.
 - Adjustment inside a First Order Loop.

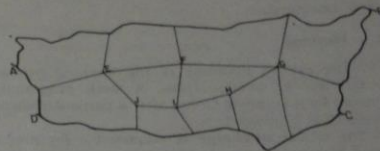


Fig. 2

- Third Order:**
- i.e. sub-division of Second Order
 - No geodetic corrections (the earth's surface is considered as a plane).
 - Adjustment inside the Second Order loop.

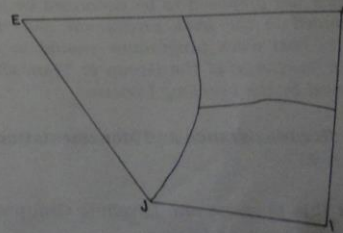


Fig. 3.

4. WORKING PROCEDURE

The work of the Levelling Division can be divided into the following main headings.

4. 1. Planning
4. 2. Reconnaissance and Monumentation
4. 3. Observation
4. 4. Computation

4. 1. Planning

The overall planning of the levelling network is carried out at headquarters. A work programme is prepared for a Levelling Group after a particular alignment is finalized at headquarters.

The officer in-charge is responsible for preparing detailed technical work plan for the execution of the task assigned to him. In addition to this, a chart in the scale of 1:50,000 or equivalent should be prepared, showing the proposed route of the levelling line together with all the existing Geodetic Levelling lines in the area with their alignment numbers and field season. The chart should also indicate the starting and closing Bench Marks and all the Permanent Bench Marks and Base Points proposed to be established in the new line. A list of Triangulation Stations which are proposed to be connected by levelling may be included in the work programme and marked on the chart. This work programme should be carried by the officer in-charge of the Group or Team after it has been approved by the Levelling Division.

4. 2. Reconnaissance and Monumentation (see chapter 5 & 6)

For this purpose one Levelling Group is divided into four working teams as follows.

Team 1: Responsible for the monumentation of Permanent Bench Marks and their description cards.

Team 2: Responsible for the reconnaissance of the location of Temporary Bench Marks and their markings.

Team 3: Responsible for the drilling of holes for Temporary Bench Marks and plotting of the alignment and location of PBMs on a 1:50,000 scale map. (or any large scale map available)

Team 4: Responsible for the monumentation of TBMs numbering, sketching and measuring the bearing.

The quality of monumentation and description cards must be checked frequently by the Team Leader or the Group Leader.

The observers will be interchanged from one team to another team in rotation.

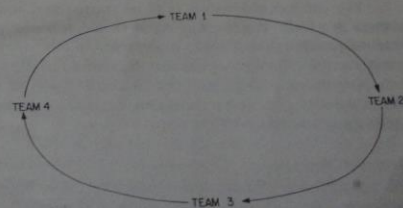


Fig. 4.

4.3. Observation (for detail see chapter 9)

For this purpose the group is divided into two teams as follows.

Team I: For Forward Levelling.

Team II: For Backward Levelling.

Note:- In surveying, one measurement has no meaning unless there is a second measurement to support it. That is why double levelling is carried out (Forward and Backward).

Each team consists of:-

- | | |
|-------------------------------------|---------|
| 1. Assistant team leader (observer) | 1 |
| 2. Booker | 1 |
| 3. Staff supervisor | 1 |
| 4. Recce-supervisor | 1 |

4.3.1. The work of a team:

The process of levelling is to be started and ended **always** at a PBM. The levelling route should run over the shortest possible distance with a minimum number of set-ups of the instruments to minimise the errors resulting at each instrument set-up. In order to avoid observations when shimmering is at its maximum, each day's work is divided into two sessions.

Observation Periods:

1. Morning session : From sunrise to approx. 10.00 hours.
2. Afternoon session : From 14.00 hours to Sunset.

The observation sub-team consists of:

Field personnel

Observer	 1
Booker	 1
Staff Supervisor	 1

and the required number of chainmen for the mobilization of the survey equipments during the observation.

Note: Always the same staff man carries the same staff (so that the observer during a reading always knows where staff A or B is located). Take care to attach always the Same thermometers to the same staff.

Preparations before the observations:

Immediately after arriving at the place of work, the **observer** first **removes the instrument** from its Containar and places it **in a shadow** which has freely circulating air for about **half an hour**, so that it can adapt to the observation conditions.

The **staff supervisor** goes quickly along the line to check whether the marks for the instrument and staff positions established by recce sub-team are still visible or not. At the same time he also checks the condition of BMs.

The **booker** supervises the quick unloading of all necessary materials. He checks the correct attachment of the thermometers. He is responsible for the **same staff** to be set up at the BM which was set up at that BM where levelling of the previous observation period ended.

The two staff men remove the material from the top rack of the vehicle. Then everybody moves with their own material to their respective positions as shown in fig. 5.

(The positions of staff A and B can be interchanged if the levelling of the previous observation ended with staff B.)

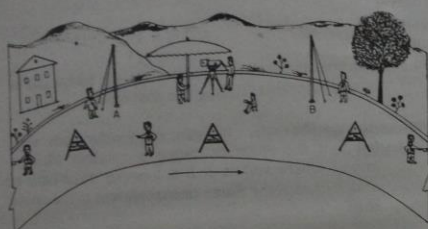


Fig. 5.

The observation is carried out by setting up the instrument and staves over the marks which were previously reconnoitred by the recce sub-team (see 5.4). The ground should be stable and more or less dry. As far as possible, the distances of the Backsight and the Foresight must be equal in order to minimize the influences of the collimation error, refraction, earth's curvature etc. (see chapter 10)

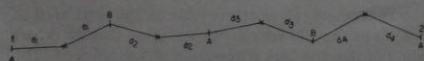


Fig. 6.

Referring to fig. 6 levelling from station 1 to station 2 is carried out by choosing several instrument positions.

The first instrument position is established at mid-point of the two staves A and B. For levelling of High Precision, staves with **double graduations** are only to be used. Thus the observation of the staves are taken as follows:

1. BACKSIGHT,
left graduations, i.e. small numbers, (a) stadia hairs
(b) centre hair
2. FORESIGHT,
left graduations small numbers, (a) centre hair
(b) stadia hairs
3. FORESIGHT,
right graduation big numbers centre hair
4. BACKSIGHT,
right graduation, big numbers centre hair

This rhythmic sequence is to be made with **equal intervals of time** between each reading. In particular the time lapse between 1 (b) and 2 (a) should be equal as near as possible to that between 3 and 4. At the next set-up the observation sequence is to be made as follows:

1. FORESIGHT, small numbers (a) stadia hairs
(b) centre hair
2. BACKSIGHT, small numbers (a) centre hair
(b) stadia hairs
3. BACKSIGHT, big numbers centre hair
4. FORESIGHT, big numbers centre hair

As previously the time lapse between the centre hair reading in small and big numbers must be kept equal.

Note:

1. The only difference in the two sequences given above is that once the backward staff is aimed first. And on

the next set-up the forward staff is aimed first, i. e. read always the same staff ('A') first.

2. Take the observations as quickly as possible one after another, but keeping in mind the time lapses mentioned above.
3. Move as quickly as possible from one instrument position to the next one and then start the observations as soon as possible. Otherwise the forward staff will have too much time lapse, during which its height may have changed without being noticed.
4. During changing from one set-up to the next, the foresight staff (change point) should be turned with extreme care so that it does not change its position.

The above mentioned sequences of observations and their speedy execution, eliminates most of the error introduced by the vertical movement of the staves.

Performing the observations in such a way, a minimum of 6 to 8 sub-sections can be levelled in one day (during the two observation periods).

4.3.2. The work of the group:

Levelling of a line is to be carried out in two directions. One in the forward direction and the other in the opposite direction, for the investigation of gross error and for the increment of levelling accuracy.

Therefore the Backward Levelling must be done as soon as possible after the Forward Levelling or vice versa.

For this reason the whole line of levelling is divided into sections of such length, so that each section can be levelled within approximately 14 days.

4.3.2.1. 14-days section (14-ds)

They always start and end at a triple or quadruple group of PBMs and never at a single BM.

As far as possible the camp is to be established in the middle of each 14-ds.

The number of sections and sub-sections in a 14-ds depend on the nature of the terrain. This number is to be decided upon before starting the levelling of that 14-ds. This is easily done in the following way:

In one day a minimum of 8 sub-sections can be levelled, i.e. in 14 days a minimum of 112 sub-sections can be levelled.

At the time of reconnaissance, a list of Bench Marks (Lev. Form 1.24) are prepared (see chapter 5). From the triple or quadruple group at the end of the last levelled 14-ds count in that list the 112 sub-sections in the direction of levelling. Now take the nearest triple group as the end of the new 14-ds.

The two teams level the 14-ds from opposite directions.

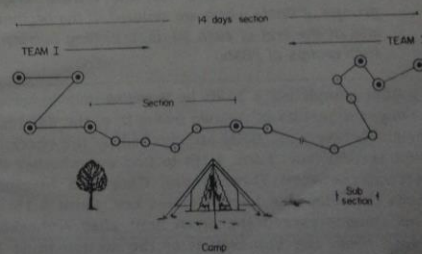


Fig. 7.

Note: The Forward and Backward Levellings are always carried out by a different observer and different equipments. The observations are carried out as shown in fig. 7.

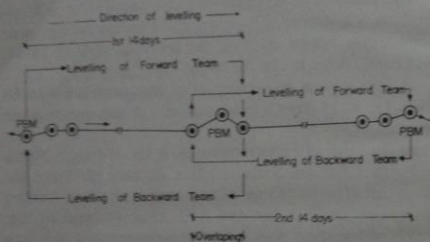


Fig. 8.

Note: Each team levels twice over the **Whole triple groups** of PBMs which are situated at the beginning and at the end of each 14-ds, i. e. they overlap at such groups of PBMs.

After completing a 14-ds by Forward and Backward Levelling, the results of the two teams (i. e. the difference of height between the BMs) are checked by the checking of that section (Lev. Form 1.16) to find out whether the differences between Forward and Backward Levelling exceeds the allowable tolerance (see Lev. Form 1.15). If the differences exceed the given limit after rechecking the field book, the sub-section or the section must be repeated. Once everything is satisfactory and within its limit, the next 14-ds is observed.

4.3.2.2. Check Levelling

If a line of levelling starts and closes on the Permanent Bench Marks, which had been previously established and levelled with a definite accuracy, the stability of such Bench Marks should be verified by check levelling. The check levelling is carried out at the group of 3 to 4 existing Permanent Bench Marks and the result should agree within the permissible limit as given in Appendix G.

The Bench Marks for check levelling should be very carefully selected so that these are adequately spaced to safe guard against the uniform local movement (sinking & upheaval).

This check levelling should also be carried out whenever the levelling line crosses the old one. During the check levelling, the description cards of the old Bench Marks should be updated if required.

The abstract of the check levelling is prepared in duplicate and a copy should be sent to head quarter for approval.

ABSTRACT OF CHECK LEVELLING

ALIGNMENT NO.				BENCH MARK OF OLD LEVELLING		OBSERVED DIFFERENCE OF HEIGHT		TOL.	REMARKS
NO.	TYPE	CONDITION	DATE	DIST. (km)	DATE	OLD LEVELLING (cm)	NEW LEVELLING (cm)		
						DIFF. (cm)	DIFF. (cm)		

4.4. Computations

This instruction book deals mainly with field work, so the computation aspect is not described here. For field plotting and checking see chapter 13.

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5. RECONNAISSANCE

Reconnaissance is the basic and most important step in all surveying work. The importance of this step is sometimes overlooked by the observer, but the utility of a thorough reconnaissance cannot be overemphasised, the time spent being amply repaid in the greater ease with which the levelling can be executed.

Generally reconnaissance is done in three stages:

- (a) choosing the best suitable alignment on the basis of the available topographical and geological maps of the area. This stage is very important and it is advisable to consult the geologist for this purpose. Normally this stage of reconnaissance is carried out at headquarters.
- (b) The determination of the **exact location of each PBM and TBM** in the field is of vital importance. Especially for DS and BP it is advisable to have a geologist on the site.
- (c) The determination of the exact locations of the instrument and the staves.

This third step is to be carried out during observations, but the first two stages are to be carried out **before** (and not during) the **monumentation** work of the concerned line.

5.1. Alignments

Suppose the levelling between any two given points A and B could be done by different alignments but only one is needed, in such cases the suitable alignment can be selected keeping in view the following conditions. The

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marks are allocated in order of priority, thus the alignment which scores higher marks is the best suitable alignment.

Conditions	Marks for		
	1st	2nd	3rd
	order		
- The alignment proceeds through geologically stable areas	9	9	3
- Instruments and staves to be set on stable ground	8	8	9
- Traffic does not affect the stability of staves and instruments by vibration.	7	7	7
- Good rock for monumentation is available all along the alignment.	6	6	4
- A minimum of BMs proposed to be monumented by pillars	5	4	8
- The alignment can be levelled by a minimum of instrument set-ups.*	4	3	5
- The road construction is stable	3	2	2
- The alignment proceeds through an area where BMs are needed.	2	5	6
- The levelling team can move along the alignment by vehicle.	1	1	1

* The number of necessary instrument set-ups for the levelling of an alignment depends on the length L (in km) and the gradient g (in %) of the alignment and on the total of absolute differences of heights $\sum |\Delta h|$

$$\sum |\Delta h| = |\Delta h_1| + |\Delta h_2| + |\Delta h_3| + \dots + |\Delta h_n|$$

$$g = \frac{\Delta h \text{ (in metres)}}{10 \times L \text{ (in km)}} \% \quad (5.1.a)$$

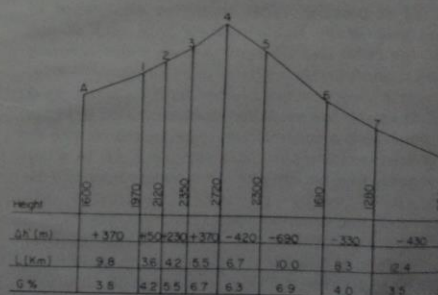
If $g \leq 4\%$ $n_1 = 20 \times L$ (5.1. b.)
 $\therefore$ max. line of sight = 25 m. Thus 1 km needs 20 set-ups.

If $g \geq 4\%$ $n_2 = \Delta h / 2$ (5.1.c)
 $\therefore$ In one set-up max. of 2 m Δh can be levelled.

So the number of necessary set-ups to level an alignment becomes

$$n = n_1 + n_2 \quad (5.1.d)$$

Example: (Profile of the alignment not to scale.) The height and distances are to be taken from a suitable map.



From A to 1 and from 6 to B $g \leq 4\%$.

Therefore $n_1 = (9.8+8.3+12.4) \times 20 = 610$

From 1 to 6 g > 4%

$$\begin{aligned}\text{Therefore } \sum \Delta h &= |+150| + |+230| + |+370| + |-420| + |-690| \\ &= 150 + 230 + 370 + 420 + 690 \\ &= 1860 \\ n_2 &= 1860 / 2 = 930 \\ n &= n_1 + n_2 = 1540.\end{aligned}$$

Thus a levelling traverse from A to B requires at least 1540 set-ups. From this, one can calculate the time required for the completion of the observation.

5.2. Permanent Bench Marks

The approximate distance between two PBMs is said to be 2 km but it can be longer or shorter (1.5 km to 2.5 km) depending on the existence of suitable places for monumentations.

As far as possible, PBMs should be established in hard and geologically stable rock. If hard rock is not available, the PBMs can be established on stable concrete structures such as bridge supports, culverts etc.

If over a long distance neither hard rock nor good concrete structures are available, monumentation of PBMs may be done with a concrete structure casted on site or with a precast concrete pillar at a safe place, i.e. in a dry field (but never in a paddy-field). Such pillars may only be established if really no other possibility for monumentation can be found, and in flat areas only.

PBMs are always established out of the reach of traffic. Therefore, if possible, PBMs should be established at about 50 to 100 metres away from the road on a safe accessible place.

PBMs may not be established near big trees, as growing roots may disturb their stability. Always keep in

mind, that it must be possible to set up the staff on the BM, i.e. the area 2 metres around and at least 3.20 metres above the BM must be clear.

If PBMs are established as a group, the BMs are to be established close together, so that the difference of height between two PBMs (or between the TBM and PBM) can be determined with two to four instrument set-ups only. During reconnaissance a sketch has to be drawn which shows the location of reconnoitred BMs along the alignment and the proposed location of the PBMs are marked with enamel paint.

5.3. Temporary Bench Marks

The reconnaissance of their locations is carried out after the reconnaissance of PBMs.

TBMs are established always where PBMs are more than one length of sight from the road. If there is a group of PBMs the TBM is established in the centre of that group very close to the road.

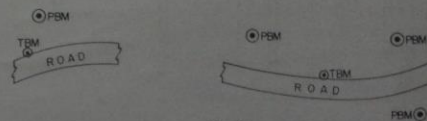


Fig. 9.

TBMs are established so close together along the alignment, that the difference of height between two consecutive TBMs (or between a PBM, which is situated very close to the road and a TBM respectively) can be levelled with 6 to 8 instrument set-ups, i.e. within approximately one hour.

It is obvious, that in flat areas the intervals between TBM's are longer than in the slope areas. The length of such intervals is determined by the following procedure.

Two sticks, one for the instrument height (1.50m) and another for the staff (3.20m), are prepared. The staff stick is marked at two places, i.e. 0.80 and 2.50m above ground level. Two persons are required for this job, one as observer and the other as staff-man.

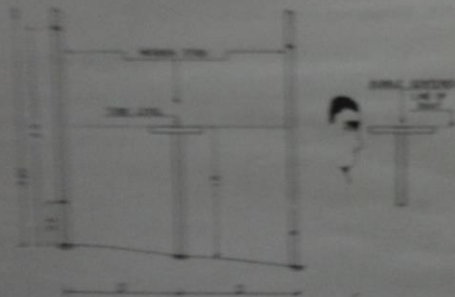


Fig. 10.

1. The staff-man with his stick stands at the place of a TBM or a previously reconstituted TBM.
2. The observer with his stick and tube-level paces of the distance (maximum 25m) from the staff position

and locates a suitable position for the instrument, so that his line of sight, i.e. the horizontal line of the tube-level over the 1.50m stick is between the two marks of the staff-stick. If it is OK, he signals to the staff-man "O.K.", if not, he changes his position accordingly.

3. The staff-man after receiving the signal, paces the direct distance from his staff position to the instrument i.e. D in the figure above and paces exactly the same distance from the instrument towards the new staff position.
4. After the staff is set in the new position, the observer checks whether his line of sight lies between the two marks of the staff or not. If not, the positions of the staff and of the instrument are changed accordingly. When it is O.K., the observer counts the number of instrument set-ups and changes his instrument position to the new position.
5. After executing this procedure six times i.e. six instrument positions are set out, one traces the place where the location of the next TBM is to be looked for.

TBM's are always established very close to the road on a safe place.

They are established in rock or good concrete structures wherever is available on the ground.

Check always that it is possible to set up the cases at the reconstituted places. The reconstituted situations of TBM's are to be marked with paint.

During maintenance a sketch has to be drawn which shows the approximate location of reconstituted TBM's along the alignment.

5.4. Reconnaissance of the positions of staves and instrument

This work is of great importance for the execution of the High Precision Levelling and therefore it must be carried out with great care. This reconnaissance must be carried out at least one day prior to the observations. The aim of this work is to expedite the observations so that a minimum time is taken in connecting one Bench Mark to another. Thus the observations can be run smoothly.

For this reason the reconnaissance sub-team has to perform the following tasks.

1. Uncover and check the conditions of all BMs.
2. Remonumentation of destroyed BMs if any, well in advance before the observations.
3. Reconnaissance of the minimum of necessary staff and instrument positions between two consecutive BMs making sure that there are always an even number of set-ups.
4. Foresight and Backsight for each set-up must be of equal length.
5. The positions for the staves and the instrument must be chosen on stable ground, but if possible not on asphalt.
Best positions are on
 - fine gravel or sand.
 - concrete surface.
 - big, embedded stones.
6. If there is no other way than to proceed over grass-land, the grass-layer is dug out and the instrument/staff is placed over it.
7. All reconnoitred positions are marked in such a way that the marking must last until the time of observation. It is done either by pick or by paint.

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- Material required - 30m nylon rope.
- 1.5m long wooden stick (instrument).
 - 3.20 m long wooden stick (staff) with one mark 0.50m and another mark 2.50m from the bottom of the stick.
 - Tube level.
 - Marking paint and brush.
 - Pick.

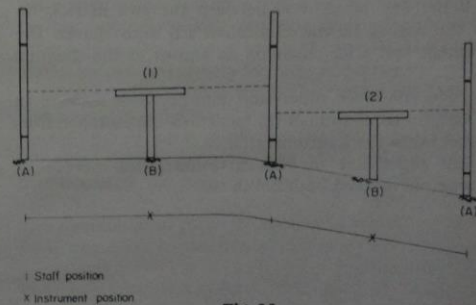


Fig. 11.

Operation:

One chainman (A) with the staff and one end of the rope erects his "staff" over the TBM or PBM, the reconnaissance supervisor with instrument stick and tube level locates the suitable position for the instrument and checks the staff (line of sight must be between the two marks of the staff). When he gives the signal, the other chainman (B) measures the distance of the staff-

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instrument by marking the other end of the rope, the chainman (A) with the staff and one end of the rod proceeds forward till his rope is tight and chooses a suitable (i.e. stable) place for the staff.

Note: The selection of stable staff positions as well as the equality of Backsight and Foresight are of utmost importance.

The more-supervisor advises the chainman in selecting the stable staff positions. When the position is chosen, he checks the staff reading from the instrument position (1).

If the line of sight is between the two marks, he gives the signal to the chainman (B) who marks the instrument and staff positions as shown in the diagram above.

The more-supervisor then proceeds forward to position (2) and chainman (B) measures the distance from the staff (A) to the instrument (2).

By repetition of this procedure all necessary positions are marked from BM to BM.

6. MONUMENTATION

Monumentation is the process, by which the levelling station markers called "Bench Marks" are established in a reconnoitred place (see. reconnaissance). The Bench Marks thus established should remain intact for ever, thus extreme care must be taken while carrying out this monumentation. For this reason, the Bench Marks are established in a safe and protected area and in the case of a Base Point & Datum Station, a protection wall or railing should be constructed along with a marble/brass plate indicating its name and year of monumentation. During monumentation one should explain to spectators in a friendly and polite way about the importance of Bench Marks so that it will help for their protection.

All monumentation should be carried out just before and during the monsoon season, if possible at least one season in advance. However the following time gap between monumentation and observation must be maintained for the stability of the Bench Marks.

Monumentation in rock: at least two weeks.

Monumentation in concrete structures: at least two months.

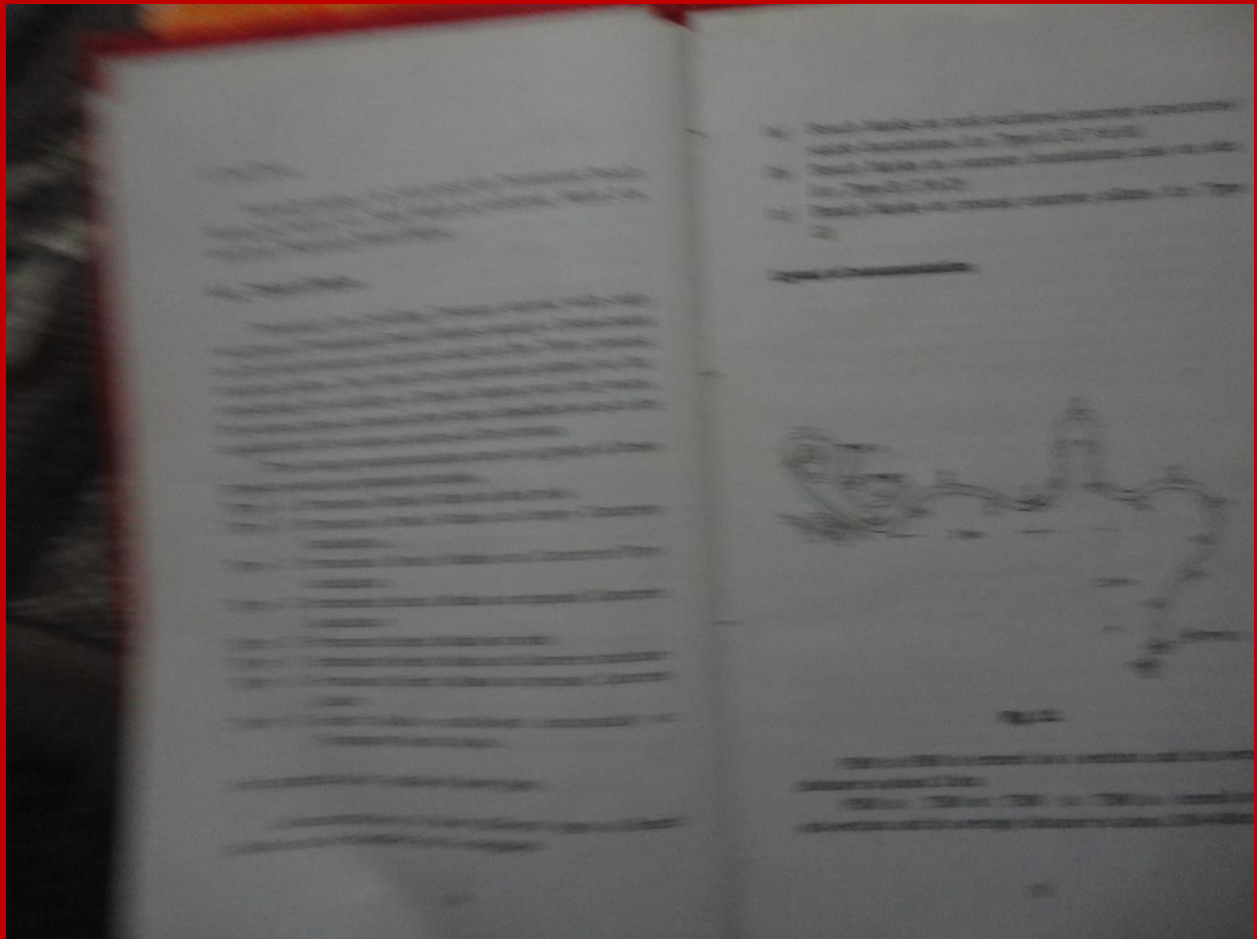
Monumentation with concrete pillars: at least six months.

Monumentation of Datum Stations or Base Points in a concrete foundation: at least one year.

6.1. Kinds of marks

There are various kinds of marks used for the Bench Marks.

1. Brass mark
2. Steel mark



3.3. Bench Marks on rock surfaces/concrete structures/stable foundations

1. Prepare the surface of the rock/concrete by cutting a cubical hole of (15 x 15 x 10)cm.
2. In the centre of that area, drill a vertical hole. This hole should be as narrow as possible, i.e. of the same diameter as that of the kind of mark being used. It must be at a depth that only the head of the mark (1 cm of the steel/brass-mark or rivet) is exposed out of the hole, i.e. the base of the marks are supported by rock (or concrete in concrete structures) but not by filled in cement.
3. All dust and soil etc. must be removed from the hole by water i.e. the hole is washed out thoroughly.
4. Remove the water completely from the hole. Sprinkle some cement into the hole to dry up the water particles.
5. The mortar for the concrete consists of:-
Clean sand (without any organic ingredients) mixed 1:1 with cement and clean water. The concrete must not be too soggy.
6. The mark is placed in the hole and the remaining cavity is fully filled in with concrete. Make sure that the mark rests on the base of the hole by slightly tapping the mark down. The compaction of the concrete is carried out with a thin stick or rod. Be careful not to destroy the smooth spherical surface of the mark.

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7. Make sure, that the staff is supported by the Bench Mark and not by surrounding rock/concrete.
8. Cover the BM with damp soil or something similar to prevent it from drying up quickly.

Example of good monumentation

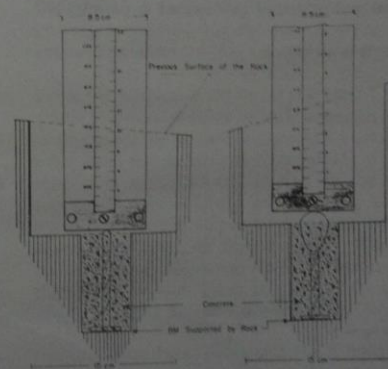


Fig. 13

9. After the monumentation of the mark, the important stage is its protection. The surface of the Bench Mark is covered by a Bench Mark Cover or by some other arrangement so that the surface remains intact. The hole is then filled in with sand or soil and

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the upper surface of the rock/concrete structure is plastered to its original level.

10. A good description card will help to relocate the position of the mark and thus by breaking the plaster surface, the Bench Mark can be exposed for the observation.
11. One should make sure that the Bench Mark exposed during observation or found to be exposed, must be covered and plastered as described in para 9.

A bad example is as follows

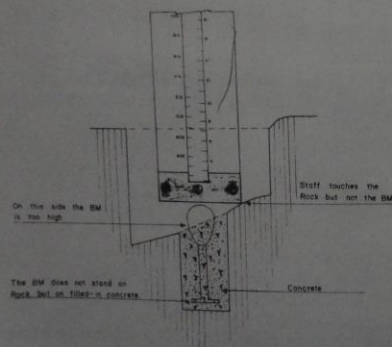


Fig. 14

6.4. Bench Mark on a concrete foundation cast on site

For the monementation of these types of Bench Mark, the excavation of the earthwork and the concrete foundations are carried out as per the specification of those types of marks specified in appendix A. Para 9 to 11 of 6.3 are also applicable to this.

6.5. Bench Mark on precast concrete pillars

1. Dig a hole not larger than 80 x 80 cm. square. it must be deep enough so that the surface of the pillar is 30 cm below ground.
2. Level the bottom of the hole and ram it with some gravel or pieces of brick in such a way that the pillar is supported by a compact rammed surface. i.e. the pillar must not stand in the loose soil.
3. Place the pillar in the centre of the base and surround it completely with stones. Press these stones by hammering each against the another and against the pillar, so that the pillar does not move laterally. Cover the stones by an approximately 10cm thick layer of soil. Press the soil down thoroughly. Ensure that the pillar stands vertically.
4. Place in again 10cm soil and press it down. Thus the soil is compacted in the hole by slowly pressing down several times.
5. Repeat step 4 until approximately the upper 15cm of the pillar is visible.
6. Surround the pillar again with stones and proceed as in para 3.

7. After finishing the dimension description of that pillar, the BM is capped by a cover and the pillar is covered with a layer of approximately 10cm soil, without any stones. Press it down gently, but not too much. Then put a flat stone in the centre of the hole (i.e. above the mark) to protect the mark. Top up the hole fully with soil up to ground level and press it down firmly.

Example

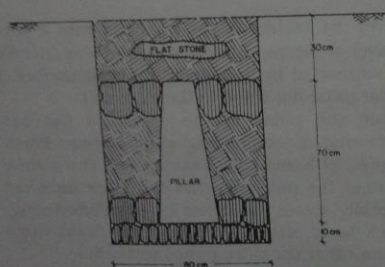


Fig. 15

6.6. Temporary Bench Mark (Iron Rivet) in pipe

This type of monumentation for a TBM is only carried out if there is absolutely no rock or concrete structure in the surrounding area.

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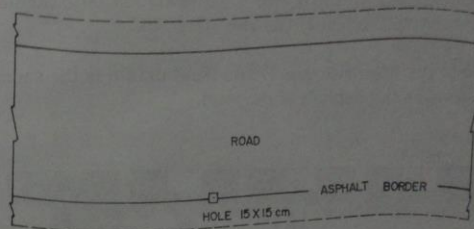


Fig. 16

1. Dig a hole at about 15cm square at the edge of the asphalt road.
2. After removing the asphalt layer, a long chisel is driven into the ground in the centre of the hole.
3. Judging the nature of the ground, choose either a 30 cm or 60cm long pipe (diameter approximately 2.5cm).
4. The chisel is removed from the hole and the hole is partly filled with small gravel. Then the pipe is driven into the hole by hammering down, using a piece of wood held over the pipe to prevent damage till it is approximately 10cm below the surface of the asphalt.
5. The pipe is then filled with gravel and it is compacted thoroughly leaving space for the iron rivet only.

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3. Fit the iron rivet into the pipe and take the dimension description, i.e., some measurements which will identify the rivet later on.
7. Fill the hole with gravel and sand until it is the same level as the asphalt of the road.

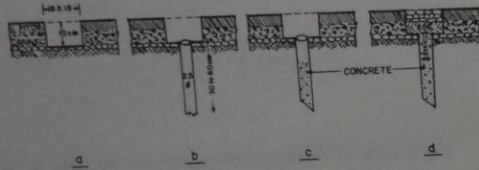


Fig. 17

7. NUMBERING OF BENCH MARKS

Correct numbering of each Bench Mark is as essential as a correct reading of the staves: Especially, one number is allocated to one single BM only, i.e. each BM has a separate number.

Therefore, during the time of numbering in the field, a list of allocated numbers is to be sent monthly to headquarters for registration.

7.1 Fundamental Bench Marks, Datum Stations and Base Points

Such Bench Marks are characterized by the name of the place where they are situated (i.e. name of the respective town, village, temple, etc.). Such stations are established by a single BM only, to avoid confusion. However, to be able to control the stability of such BMs (and eventually to restore them) at least two other types of BMs are established near them. But these additional BMs are either treated as Base Points or as first order PBMs within that line.

7.2 First Order Levelling (High Precision Levelling)

While establishing a levelling network in a country, levelling is carried out from one known reference point to another unknown point following a line. Thus the numbering of Bench Marks is adopted in terms of alignment or line number.

7.2.1. Alignment Number

The First Order alignment/line number is designated by 3 digits and the starting digit is number 1 indicating

e.g.	Alignment	Number
	Birganj - Kathmandu	100
	Naubise - Mugling - Pokhara	101
	Pokhara - Butwal - Bhairahawa	102
	Butwal - Hetauda	103
	Pathalैया - Kankadbhitta	104

Three digits are assigned (i.e. from 001 to 999) for the numbering of Permanent Bench Marks. The numbering is carried out continuously for one alignment in the direction the levelling proceeds (Forward Levelling). This makes it possible to number maximum of 999 Bench Marks in one alignment.

e.g.
103 - 050 1 - representing order of work
 103 - alignment/line number with 3 digits
 050 - Bench Mark number in the
 direction of Forward Levelling.

Since this system of numbering alignments restricts up to 100 different alignments only, one has to be careful

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First Order PBMs are established alternately as single points and as groups of three to four, the distance between a group and a single being approximately 2km.

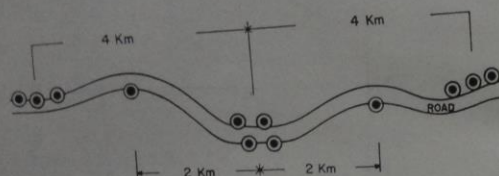


Fig. 20

From the sketch above it can be seen that over a distance of 4 km an average 4.5 PBMs are required for a distance of 1 km, an average $(4.5/4)$ are required. For a distance of L km the total number T of PBMs required is

$$T = (4.5/4) \times L$$

$$= (9/8)L \dots\dots\dots (7.2.2)$$

Thus, for a distance of 200 km, the total number of First Order PBMs required are:-

$$T = 200 \times 9/8 = 225$$

7.2.2.1. Numbering of destroyed or additional PBMs

If some PBMs are found destroyed prior to the observation, such PBMs can be reestablished close to the previous PBM positions. These PBMs will carry the same previous numbers but the old description cards should be superseded by the new ones.

If some PBMs are found destroyed or missing after the observation, the number of such PBMs are cancelled from the list and a remark "Destroyed" is noted on its description card.

It may happen that the destroyed PBM positions need to be replaced by a new PBM or some additional PBMs need to be established between the existing old PBMs. In such cases the numbering of the new PBMs will be carried out as follows.

Example:-

Alignment No. 103

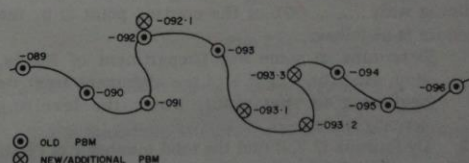


Fig. 21

Let us assume that one PBM needs to be established instead of 103-092, which was destroyed after observation. The new PBM number will carry the same old number and one extra digit after the decimal point.

i.e. 103 - 092.1

It may be required to have some additional PBMs between the existing old PBMs say between 103 - 093 and 103-094, for the densification of Bench Marks. Such new PBM numbers will carry the preceding PBM number and one extra digit after the decimal point. This extra digit will have the number 1 to 9 depending on how many additional PBMs are established.

i.e. 103 - 093.1
 103 - 093.2
 103 - 093.3

This system enables one to establish 9 additional PBMs between the old PBMs.

i.e. in 2 km (length between the PBMs)

There is a possibility of establishing an additional 9 PBMs at every 200 m interval.

7.2.2.2 Intermediate Numbering

It sometimes happens that levelling of an alignment does not start at the very beginning of that alignment. To have later on a continuous numbering for each alignment beginning with 001 at the starting point of it, the procedure is as follows:

1. Determine in some way (Department of Roads, Map, etc.) the length L of the alignment from its beginning to that point, where the levelling actually starts.
2. By formula (7.2.2) find the total number T of PBMs required for this part of the alignment.
3. The starting PBM number will be $N = (T+1)$

Thus the numbering is carried out continuously from N onwards.

In doing so there may be few spare numbers left over, which can be noted and recorded.

The lines connecting the junction points C, E, D, B are observed. To complete the loop $CEDBC$, it was decided to observe the line BC of the alignment 119 before the line AB of that alignment. B is already numbered as 118 - 237.

$AB = L = 160$ km

Total number of PBMs (T) required for the line AB

$$T = (9/8) \times 160 = 180.$$

Therefore the starting PBM number for the line $BC = 119 - 181$

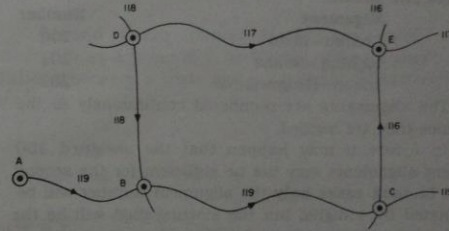


Fig. 22

7.2.2.3 Maximum length of an alignment

The maximum numbers of PBMs in one alignment is

$$T_{\max} = 999$$

$$T = (9/8) \times L \quad \text{..... (from 7.2.2)}$$

$$L_{\max} = (8/9) T_{\max}$$

$$= (8/9) \times 999 \text{ km} = 888 \text{ km}$$

Thus while planning an alignment it can be no longer than 880 km.

7.3 Second Order Levelling

As in the First Order, the numbering of Bench Marks is adopted in terms of alignment/line number.

7.3.1 Alignment Number

The Second Order alignment/line number, is initially designated by 3 digits and the starting digit is number 2

indicating the order of work. This enables one to number 100 different alignments, i.e. 200 to 299. After assigning the alignment number of the Second Order lines, a list is prepared and issued.

e.g.	Alignment	Number
	Charali - Illam	200
	Rajbiraj - Siraha	201
	Lahan - Bhagawanpur	202

The alignments are numbered continuously in the sequence they are levelled.

In future it may happen that the assigned 100 different alignments may not be sufficient for the second order. **In such cases only** the alignment number will be designated by 4 digits, but the starting digit will be the same (i.e. 2 indicating the order of work).

This will enable one to number an extra 1000 alignments

e.g.	Last alignment number in	3 digits	299
	first " "	4 digits	2000
	last " "	4 digits	2999

7.3.2 Numbering of PBMs

The Permanent Bench Marks are numbered with three digits (from 001 to 999) continuously for each separate alignment in the direction the levelling proceeds (Forward Levelling).

Thus the final number of Second Order PBMs consists of three digits for alignment and for PBMs as well.

e.g.	201 - 098	2	- Representing order of work
	201	Alignment/line number with 3 digits.	
	098	- Bench Mark number in the direction of Forward Levelling.	

201 - 098 - denotes the PBM no 98 of the Second Order alignment/line between Rajbiraj and Siraha (Al. No. 201).

Second Order Bench Marks are established alternately as a group of three (Triple) and as a single, the distance between a triple and a single is approximately 2 km.

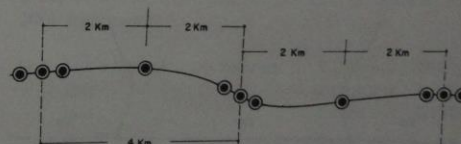


Fig. 23

From the sketch above it is clearly seen that the total number of Bench Marks required (T) is as in the first order.

e.g. $T = (9/8) L$ (see 7.2.2)

7.3.2.1 Numbering of destroyed or additional PBMs

The process is the same as in the First Order. (see 7.2.2.1)

Example:- Numbering of additional PBMs between the old PBMs.

218 - 055 and 218 - 056 are as follows.

218 - 055.1

218 - 055.2

218 - 055.3

indicating the order of work. This enables one to number 100 different alignments, i.e. 200 to 299. After assigning the alignment number of the Second Order lines, a list is prepared and issued.

e.g.	Alignment	Number
	Charali - Illam	200
	Rajbiraj - Siraha	201
	Lahan - Bhagawanpur	202

The alignments are numbered continuously in the sequence they are levelled.

In future it may happen that the assigned 100 different alignments may not be sufficient for the second order. In such cases only the alignment number will be designated by 4 digits, but the starting digit will be the same (i.e. 2 indicating the order of work).

This will enable one to number an extra 1000 alignments

e.g.	Last alignment number in	3 digits	299
	first " "	4 digits	2000
	last " "	4 digits	2999

7.3.2 Numbering of PBMs

The Permanent Bench Marks are numbered with three digits (from 001 to 999) continuously for each separate alignment in the direction the levelling proceeds (Forward Levelling).

Thus the final number of Second Order PBMs consists of three digits for alignment and for PBMs as well.

e.g.	201 - 098	2	- Representing order of work
	201	Alignment/line number with 3 digits.	
	098	- Bench Mark number in the direction of Forward Levelling.	

201 - 098 - denotes the PBM no 98 of the Second Order alignment/line between Rajbiraj and Siraha (Al. No. 201).

Second Order Bench Marks are established alternately as a group of three (Triple) and as a single, the distance between a triple and a single is approximately 2 km.

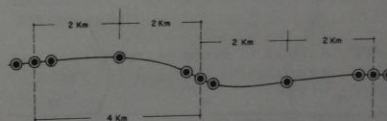


Fig. 23

From the sketch above it is clearly seen that the total number of Bench Marks required (T) is as in the first order.

e.g. $T = (9/8) L$ (see 7.2.2)

7.3.2.1 Numbering of destroyed or additional PBMs

The process is the same as in the First Order. (see 7.2.2.1)

Example:- Numbering of additional PBMs between the old PBMs.

218 - 055 and 218 - 056 are as follows.

218 - 055.1
218 - 055.2
218 - 055.3

7.3.2.2 Intermediate numbering

The process is the same as in the First Order. (see 7.2.2.2.)
Example:-

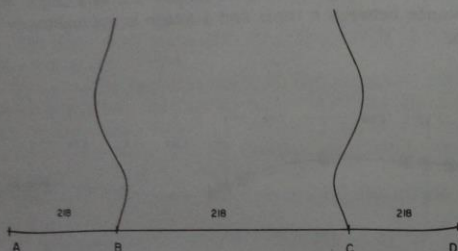


Fig. 24

Alignment No. of ABCD = 218
Length AB = $L = 168$ km
Total number of PBMs required = $T = (9/8) L = 189$

Therefore, the starting PBM number for the section BC of the alignment = $N = T + 1$

Thus the numbering of the PBMs in the section BC is carried out continuously from 190 onwards.

7.4 Third Order Levelling

In this case as well the numbering of Bench Marks is adopted in terms of alignment/line number.

7.4.1 Alignment

The Third Order alignment/line number is initially designated by 3 digits and the starting digit is number 3, indicating the order of work. This enables one to number **100 different** alignments i.e. 300 to 399.

As in the case of the Second Order, the assigned 100 different alignments may not be sufficient. **In such cases only** the alignment number will be designated by 4 digits but the starting digit will be the same (3 indicating the order of work). This will enable one to number an extra 1000 alignments.

i.e.	last alignment number in	3 digits =	399
	first " "	4 digits =	3000
	last " "	4 digits =	3999

7.4.2 Numbering of PBMs and additional PBMs

The system and procedure is exactly the same as for the second order PBMs. (see 7.3.2 & 7.3.2.1)

Example:- 303 - 054 denotes the PBM No. 54 of the Third order alignment No. 303.

7.5 Temporary Bench Marks

The system and procedure to number them is exactly the same for First, Second and Third order levelling.

Temporary Bench Marks are established between two PBMs always after 6 to 8 instrument set-ups. The TBM numbers are given as fractional numbers, i.e. $125/1$, $125/2$. Where the numerator denotes the Bench Mark numbers from which the line started and the denominator

denotes the serial number of the Temporary Bench Marks in that line.

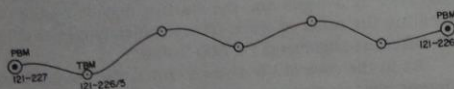


Fig. 25

It may often happen that a section does not start or end at a PBM (see the diagram), but at a TBM which is situated somewhere in a central position on the road between the triple group of the PBMs.

It is recommended to number such a central TBM with the highest PBM number in that group of PBMs. (see 121 - 222/1)

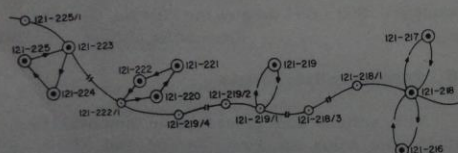


Fig. 26

If PBMs lie in the line of levelling, the following TBMs will be numbered with the highest PBM number in that group. (see 121 - 225/1).

7.6. Numbering of Staff Positions

If the staff is set up on a PBM, TBM or Trigonometrical Station (i.e. on any mounumented point) it gets the number of that particular BM or TS.

If it is set up on the ground plate it will be numbered with the number of the previous PBM or TBM number and in addition with small letters continuously.

Example:-

Staff on	PBMs	121 - 218	in short - 218
"	Ground plates	121 - 218/a	-218/a
"	"	121 - 218/b	-218/b
"	"	121 - 218/c	-218/c
		etc.	
"	TBMs	121 - 218/1	-218/1
"	Ground plates	121 - 218/1a	-218/1a
"	"	121 - 218/1b	/1b
		etc.	
"	TBMs	121 - 218/5	-218/5
"	Ground plates	121 - 218/5a	/5a
		etc.	

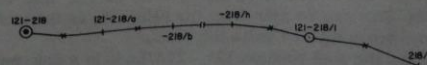


Fig. 27

7.7 Numbering of Staff Positions and TBMs in Links

For the computations in the Head Office it is essential to know precisely, which data serves to the

computation of the differences of heights between two consecutive BMs of that line of levelling and which ones are data of links.

Such link observations do not serve the main line of levelling, but always start and end at PBMs of a line. The positions of the staves in such cases will be numbered as follows:

Example:-

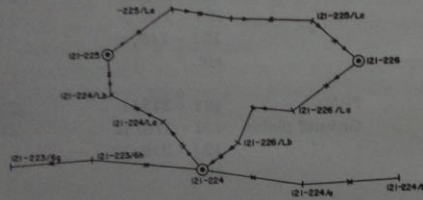


Fig. 28

Here the small loop 121 - 224 → 121 - 225 → 121 - 226 → 121 - 224 does not belong to the line of levelling. This can be seen from the fieldbook, as the staff positions are numbered as/La/Lb,

Thus the staff-position 121 - 224/La is in a link, but the staff position 121 - 224/a is in a line.

Similar is that for a TBM, if it is needed in long links:

Examples:- 121 - 227/L1 is a TBM in a link
121 - 227/L1a is a staff position in a link
121 - 227/1 is a TBM in a line
121 - 227/1a is a staff position in a line.

Note:- In the example of fig. 28 the PBM 121 - 224 is common for the line and the small loop, hence it's number will be in the fieldbook of the line and in the fieldbook of the link.

8. DESCRIPTION CARDS

The preparation of description cards are the vital part in the process of monumentation. This consists of all the necessary information and measurement of the Bench Mark locality, thus serving as an identity for that particular Bench Mark. By the help of this, the Bench Mark can be recovered any time in future.

Almost all the Bench Marks are monumented under ground level or well inside the rock surface to protect them from destruction. That is why the description cards must be neatly completed during monumentation.

8.1 Completion of Description Card

The following information must be filled in Lev. Form 1.20.

- (a) District
- (b) Panchayat
- (c) Ward No.
- (d) Grid Sheet No.
- (e) One Inch Sheet No.
- (f) Type of Bench Mark
- (h) Kind of Land
- (i) Land owner

8.1.1 Written Description

This description must be very short (in Nepali) but in detail. This must give the exact location, any prominent feature and the distance from the nearest village or town. The location of a Bench Mark **above** or **below** ground level must also be given here.

8.1.2 General Sketch

A general sketch (oriented to the north) covering a large area must be drawn to give the rough location of a Bench Mark. The prominent features such as buildings, trees, temples etc. must also be sketched. The shape of the terrain must be shown by the help of contour lines. If there are not any man made or physical features around the Bench Mark, the general sketch must be extended up to the nearest prominent feature.

For example:-

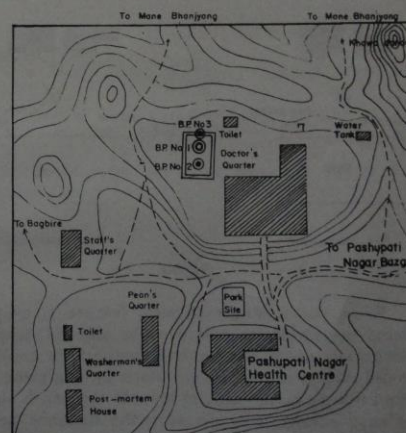


Fig. 29

8.1.3 Dimentional Description

This must be done in large scale with the proper orientation and all the possible measurement must be carried out. These measurements are carried out for the recovery of the station in the future, that is why as far as possible, only physical features must be chosen for measurements. A minimum of 3 best cutting measurements will be sufficient rather than to have several poor cutting measurement.

For example:-

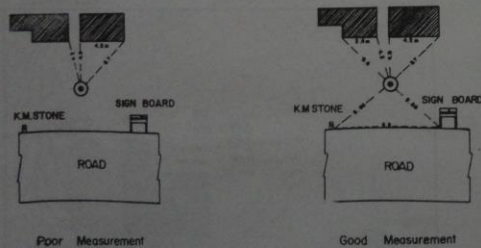


Fig. 30

When not a single physical feature is available, bearing to prominent points could also be used. The measurement could also be done from a large boulder or exposed rock near the Bench Mark by engraving a "V" shaped witness mark pointing towards the Bench Mark. It is always advisable to measure the distance between physical features (measured) or witness marks.

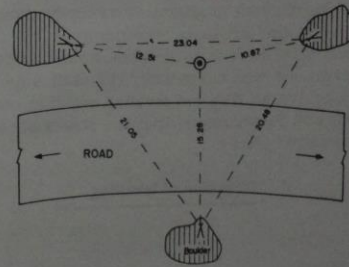


Fig. 31

All the features must be depicted as in the conventional signs.

(See Appendix C)

9. OBSERVATION

The observation procedure here is described for High Precision/First and Second Order Levelling. Before observation, the location of the BMs and the markings of staff and instrument positions (previously done by the recce supervisor) are to be reconnoitred by the staff supervisor. This recce is to be done immediately before the observations (see chapter 5). The staff supervisor is responsible, that **all BMs are levelled** and that not a single one is left out. Furthermore, that the staves are set up at the reconnoitred places.

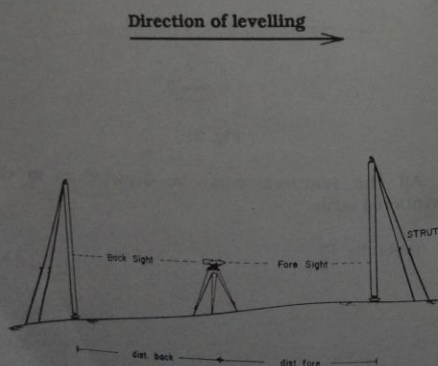


Fig. 32

The observation has to proceed very quickly without **any break** from one fixed point to another fixed point (PBM/TBM).

9.1 Operation

- (1) The staff supervisor establishes the same staff on the BM, which was set up there at the end of the previous session. The staff is set vertically by centering exactly the circular bubble attached to the staff. Now the staff is rotated one full turn and bubble is centred again. The rotation of the staff must be smooth. If not, check whether the staff touches some where else rather than the BM or the ground plate knob, clean the bottom of the staff and repeat the whole procedure. There after, the staff supervisor signals to the observer "Ready" and goes to the forward staff.
- (2) At the same time, the observer sets up the instrument insuring stability over the mark, **centres the tilting bubble** accurately and starts the readings to the backward staff as soon as the observer thinks that after these reading, the forward staff will be ready.

Before the first reading to the staff, the observer measures the height of the instrument, which is booked immediately into the field book.

- (3) The staff supervisor erects the forward staff.

*Note: While moving from one set-up to the next one, the **staff** which does not change its position, **may no more be rotated a full turn: But only the amount**, which is necessary to direct it exactly towards the new instrument position. This motion is executed **very carefully**. Centre the bubble thereafter again if required.*

- (4) The observer takes all necessary readings.
- (5) The booker computes all the necessary checks. If every thing is o.k. the booker gives the signal to move to the next set-up.

9.2 Setting up the staves

- (1) If the staff is to be erected **over a BM, the ring** on the bottom of the staff **is removed**.
- (2) **The ground plates** must be pressed with a hard stroke into the ground, so that they remain in the same position within a hundredth of a millimetre as long as the staff is set over them. They must be approximately horizontal. Their spikes are to be exchanged and repointed if they are no longer well pointed.
- (3) The bottom of the staff is first placed **near, but not on** the ground plate (or BM).
- (4) The staff is now erected approximately vertical.
- (5) The struts are placed so that they,
 - form an approximately right angle with the bottom of the staff.
 - support the staff **against the wind**
 - are not reaching too far **in the road**
 - cannot slip from their position.
- (6) The bottom of the staff is now placed **carefully** on the ground plate or on the BM, cleaning it from sand and dust.

- (7) The staff's circular bubble is now centred accurately.

(8) **Never set a ground plate on asphalt.**

- (9) If there is no choice other than to set up the staff on asphalt, the staff supervisor first hammers a roadnail into the asphalt and the staff is set on this one.
- (10) The staff supervisor removes the road-nails after use.

9.3 Setting up the instrument

- (1) Before fixing the instrument into the tripod, the tripod head has to be cleaned.
- (2) The fixing screw must not be screwed in too tightly, otherwise the footscrews will be too tight.
- (3) The instrument is set up over the mark given by the reece supervisor. Remember - if any - the corrections to equalize the Back and Foresight distances.
- (4) The legs of the tripod are set so that
 - the head of the tripod is approximately horizontal.
 - they are not based on asphalt, but on large stones, gravel, sand or concrete.
 - they are free of any tension.
 - they are not placed where the observer must stand for the observations.

- the height of the instrument is set in such a way that the observer's eye is on the same level as the eyepiece.

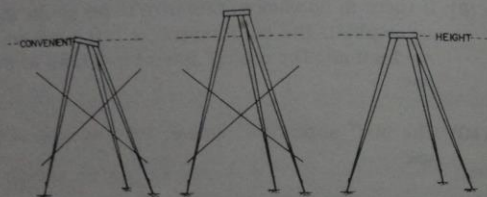


Fig. 33

- (5) The tripod is set up in such a way that the legs are pressed firmly into the ground. Be careful not to introduce any tension in the legs of the tripod in doing so.
- (6) Turn the telescope towards staff A.
- (7) Centre the circular bubble accurately, watching it in the rotatable mirror.
- (8) Start the observations.

Note: The instrument cover is always fitted over the instrument when it is not protected by the umbrella, or when carrying the instrument from one set-up to another.

9.4 Focussing the reticule of cross-hairs

A parallax-free field of view is of utmost importance for accurate observations. parallax exists if:

- (i) the reticule cross-hairs are not focussed correctly.

- (ii) the field of view is not focussed correctly.
- (iii) the observer's eye has got tired.

Note: To prevent the eyes from strain, correct focussing of the reticule cross-hairs is very important. The correct procedure is as follow:

- (1) Set up the instrument on a tripod.
- (2) Sight an object at about **10 km away** (or infinity) and focus the field of view with the focussing screw.
- (3) Hold a sheet of white paper at about 1m distance in front of the telescope.
- (4) Sighting the paper, the focussing screw must remain in its position of step 2.
- (5) The telescope eyepiece is now turned very slowly until the reticule cross-hairs appear sharp and absolutely black.
- (6) Turn the telescope eyepiece very slowly anticlockwise until the reticule cross-hairs start to go out of focus.
- (7) A very small clockwise rotation of the telescope eyepiece will again focus the hairs correctly.
- (8) Read and book the number of the **dioptric scale**.
- (9) Rotate the telescope eyepiece half a turn and repeat steps 5 to 8 again twice.

(10) The smallest and the greatest of the three readings of step 8 **may not differ by more than 1 unit of the dioptric scale**. If the difference is larger, the whole procedure is repeated.

(11) Calculate the mean of the three readings of step 8.

(12) Book the number of the instrument and the personal dioptric scale number (i.e. the mean of step 11) into the observer's note-book.

- Note:-
1. The personal dioptric scale number is always set and checked before the first reading with an instrument.
 2. The number **must not be changed** during observations.
 3. If during observation the field of view cannot be established free of parallax, check the setting of the dioptric scale. If that is correct, the observer's eye got tired.
 4. The number may be a different for another instrument and for another person.

9.5. Reading

The instrument is always protected from the sun, either by an umbrella (during reading) or by the instrument-cover (while moving from one set-up to the next).

The housing of the instrument must never be touched, but only the star shaped baseplate, footscrew, reflector knob, horizontal drivescrew, horizontal clamp, focussing knob, tilting screw and micrometer drive knob. Also take care not to touch the tripod in any way, even with your own clothing.

During observation nobody stands or walks round the instrument apart from the observer himself and the chainman with the umbrella. The booker sits near the instrument, but does not move around during the reading.

Each reading is a clearly spoken command to the booker, but not a discussion with him.

The observer after levelling the instrument accurately and setting the personal dioptric scale number starts the observations in the following sequence.

READING SEQUENCE

STAFF	BACK SIGHT				FORE SIGHT			
	STADIA	CENTRE		HAIR	CENTRE	HAIR	STADIA	
		READING	MEAN	READING		MEAN		
+	A	R 1	R 3		R 5		R 7	
		R 2	R 4		R 6		R 8	
B			R 11		R 9		R 6	
			R 12		R 10			
B		R 7	R 5		R 3		R 1	
		R 8	R 6		R 4		R 2	
++	A		R 9		R 11			
			R 10		R 12			

Fig. 34

- + Reading sequence for the 1st, 3rd, 5th set-up
 ++ Reading sequence for the 2nd, 4th, 6th set-up

Note: At each set-up, staff A (same staff) is always read first, to minimize the effect of refraction.

Fig. 35 The reticule cross-hairs

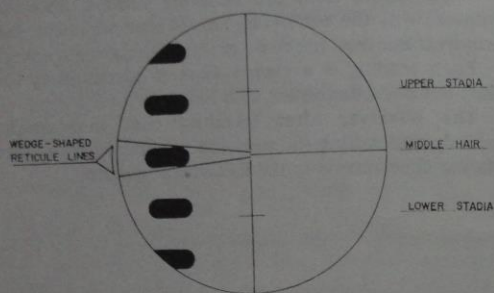


Fig. 35

As illustrated in fig. 34 the sequence of observation is as follows:

- (1) Lower stadia hair of staff A
- (2) Upper stadia hair of staff A
- (3) centre hair of staff A
- (4) Turn micrometer, set centre hair again, check bubble, read micrometer again
- (5) centre hair of staff B
- (6) 2nd micrometer reading
- (7) Lower stadia hair of staff B
- (8) Upper stadia hair of staff B
- (9) Centre hair of staff B
- (10) 2nd micrometer reading
- (11) Centre hair of staff A
- (12) 2nd micrometer reading
- (13) Rest of the information i.e. - height of instrument
- time
- temperature
- air-pressure etc.

..... small numbers

..... big numbers

Note: The time is read at each set-up immediately after the last reading to a staff, i.e. after R 12.

Remember: 4.15 p.m. is to be booked as 16.15.

9.5.1. Reading the distance, i.e. the stadia hairs

- (1) Focussing (if focussing has to be changed significantly from staff A to staff B, the observer commands the booker "beware of distance" as Foresight and Backsight distances will not be of equal length).
- (2) Centre the split - bubble.
- (3) Set the lower stadia hair by micrometer to the nearest centimeter of the **small numbers**.
- (4) Check that **no parallax** exists between the staff image and the reticule cross-hairs.
- (5) Check split-bubble. If it is no longer centred, repeat steps 2 to 5.
- (6) Calling the booker "distance" take the reading in two groups
(i) staff graduation and (ii) micrometer reading.

e.g. 151.74
Two figures only are read.
- (7) Set the upper stadia hair by micrometer to the nearest centimetre of the staff.
- (8) Check the split-bubble.

- (9) Take the reading of the upper stadia hair reading in the same way as mentioned in step 6.

9.5.2 Reading the Centre hair

The term "Centre hair" in this context always means the same as "wedge-shaped reticule lines" (see fig. 35) the procedure for the reading of the centre hair is the same for small numbers as it is for big numbers :

- (1) Centre the split-bubble.
- (2) Set the wedge-shaped reticule lines by micrometer knob to the appropriate centimetre graduation line of the staff. The wedge-shaped lines must just touch the graduation line, but they may neither lie inside nor outside of it.

Fig. 36 Correct setting of the wedge-shaped lines:

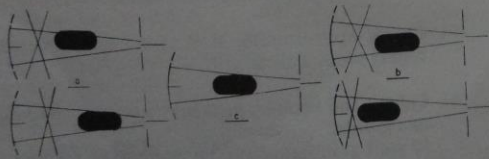


Fig. 36

- (3) Check that no parallel exists.
- (4) Check Split - bubble, if no more centred, repeat steps 1 to 4.

- (5) Check Centre hair, if no more centred, repeat steps 2 to 5.

- (6) Take the reading of the Centre hair as follows in **two groups**-

- (i) cm of the staff graduation, e.g. one four eight
- (ii) micrometer reading, e.g. six five six

148 . 656
148 . 656 cm.

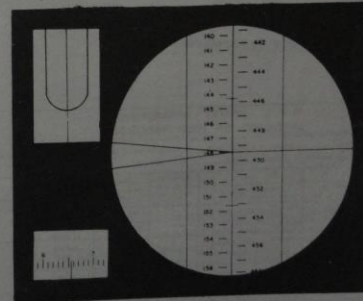


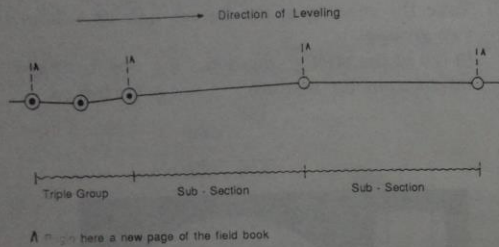
Fig. 37

- (7) Turn the micrometer half a turn.
- (8) Repeat steps 4 and 5.
- (9) Take the second micrometer reading.

Note:- On the micrometer reading the last place is estimated to only even numbers e.g. instead. 657 read 656 or 658 respectively.

9.6 Booking

For each new sub-section and for the levelling of a triple group of PBMs a new field-book page must be used.



H.M.G. SURVEY DEPARTMENT LEVELLING OF HIGH PRECISION														
POINT NO.		BACK SIGHT		FORE SIGHT		DISTANCES		HEIGHT		REMARKS				
		STADIA	CENTRE HAIR	STADIA	CENTRE HAIR	BACK	FORE	RED	WHITE					
		READING	MEAN	READING	MEAN									
1	100000	100000	100000	100000	100000									
2	100000	100000	100000	100000	100000									
3	100000	100000	100000	100000	100000									
4	100000	100000	100000	100000	100000									
5	100000	100000	100000	100000	100000									
6	100000	100000	100000	100000	100000									
7	100000	100000	100000	100000	100000									
8	100000	100000	100000	100000	100000									
9	100000	100000	100000	100000	100000									
10	100000	100000	100000	100000	100000									
11	100000	100000	100000	100000	100000									
12	100000	100000	100000	100000	100000									
13	100000	100000	100000	100000	100000									
14	100000	100000	100000	100000	100000									
15	100000	100000	100000	100000	100000									
16	100000	100000	100000	100000	100000									
17	100000	100000	100000	100000	100000									
18	100000	100000	100000	100000	100000									
19	100000	100000	100000	100000	100000									
20	100000	100000	100000	100000	100000									
21	100000	100000	100000	100000	100000									
22	100000	100000	100000	100000	100000									

Fig. 38

While booking, the booker must repeat the reading loudly so that the observer can check it to avoid

misbooking. During the observation the booker must compute and check the readings as soon as possible to inform the observer for any blunders (gross errors) before shifting the instrument.

9.6.1 Reduction and checks

(a) WHEN STAFF A IS IN BACK SIGHT

STAFF	BACK SIGHT				FORE SIGHT				DISTANCES	DIFF. OF HEIGHT
	STADIA		CENTRE HAIR		STADIA		CENTRE HAIR			
	READING	MEAN	READING	MEAN	READING	MEAN	READING	MEAN	BACK	FORE
A	R 1	R 3	R 5	R 7	R 1	R 3	R 5	R 7		
B	R 2	R 4	R 6	R 8	R 2	R 4	R 6	R 8		
C	R 3	R 5	R 7	R 9	R 3	R 5	R 7	R 9		
D	R 4	R 6	R 8	R 10	R 4	R 6	R 8	R 10		
E	R 5	R 7	R 9	R 11	R 5	R 7	R 9	R 11		
F	R 6	R 8	R 10	R 12	R 6	R 8	R 10	R 12		
G	R 7	R 9	R 11	R 13	R 7	R 9	R 11	R 13		
H	R 8	R 10	R 12	R 14	R 8	R 10	R 12	R 14		
I	R 9	R 11	R 13	R 15	R 9	R 11	R 13	R 15		
J	R 10	R 12	R 14	R 16	R 10	R 12	R 14	R 16		
K	R 11	R 13	R 15	R 17	R 11	R 13	R 15	R 17		
L	R 12	R 14	R 16	R 18	R 12	R 14	R 16	R 18		
M	R 13	R 15	R 17	R 19	R 13	R 15	R 17	R 19		
N	R 14	R 16	R 18	R 20	R 14	R 16	R 18	R 20		
O	R 15	R 17	R 19	R 21	R 15	R 17	R 19	R 21		
P	R 16	R 18	R 20	R 22	R 16	R 18	R 20	R 22		
Q	R 17	R 19	R 21	R 23	R 17	R 19	R 21	R 23		
R	R 18	R 20	R 22	R 24	R 18	R 20	R 22	R 24		
S	R 19	R 21	R 23	R 25	R 19	R 21	R 23	R 25		
T	R 20	R 22	R 24	R 26	R 20	R 22	R 24	R 26		
U	R 21	R 23	R 25	R 27	R 21	R 23	R 25	R 27		
V	R 22	R 24	R 26	R 28	R 22	R 24	R 26	R 28		
W	R 23	R 25	R 27	R 29	R 23	R 25	R 27	R 29		
X	R 24	R 26	R 28	R 30	R 24	R 26	R 28	R 30		
Y	R 25	R 27	R 29	R 31	R 25	R 27	R 29	R 31		
Z	R 26	R 28	R 30	R 32	R 26	R 28	R 30	R 32		

(b) WHEN STAFF A IS IN FORE SIGHT

STAFF	BACK SIGHT				FORE SIGHT				DISTANCES		DIFF. OF HEIGHT	
	STADIA		CENTRE HAIR		STADIA		CENTRE HAIR		V	W		
	READING	MEAN	READING	MEAN	READING	MEAN	READING	MEAN				
B	R 7	R 5		R 3		R 1						
		A 8		R 4		R 2						
A		R 9		R 11								
		R 10		R 12								
B												
C												
D												
E												
F												
G												
H												
I												
J												
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L												
M												
N												
O												
P												
Q												
R												
S												
T												
U												
V												
W												
X												
Y												
Z												

Fig. 39

Explanations to fig. 39

In the following, the numbers in brackets () represent the steps of computations as shown in fig. 39. Numbers without brackets represent numericals.

(1) $R1 - R2$

(2) Mean of $R3$ and $R4$

Check: The difference in between the two micrometer readings is $\leq \pm 0.015$ cm (15 units of micrometer).

(3) **Check:** $(1) + 2 + R2 = (2) \pm 0.05$ cm

(5) Sum of all distances in that column

(6) Mean of $R5$ and $R6$

Check: $m \leq \pm 0.015$ cm

(7) $R7 - R8$

(8) **Check:** $(7) + 2 + R8 = (6) \pm 0.05$ cm

(10) Sum of all distances in that column

(11) Mean of $R9$ and $R10$

Check: $m \leq \pm 0.015$ cm

(12) $(11) - (6)$

(13) $v = 301.550 - (12)$; Tolerance: ± 0.020 cm

This value is to be booked in units of $1/100$ mm (e.g. $0.014 = 14$). Beware of the signs (plus or minus).

(14) Mean of $R11$ and $R12$

Check: $m \leq \pm 0.015$ cm

(15) $(14) - (2)$

(16) $v = 301.550 - (15)$; Tolerance ± 0.020 cm (see (13))

(17) Upper minus lower value in that column = Δv
i.e. $v_1 - v_2 = \Delta v$; v_1 of the backward staff
 v_2 of the forward staff.

Check: $\Delta v \leq \pm 0.020$ cm
and $|v_1|, |v_2| \leq \pm 0.010$ cm.

Note: The values m , v_1 , v_2 and Δv may not be systematic in their sign, i.e. the sign should be alternately positive and negative. The values mentioned must also be within the given limits. If they are not, possible reasons for bigger differences are:

(a) If their sign, is not systematic

- The personal dioptric scale number is set wrong.
- bad focussing of the field of view.
- The observers eye has got tired.
- shimmering is too intensive; stop the work after reaching the next BM.

(b) If their sign is systematic

- if the signs of v_1 , v_2 and Δv are positive at each set-up, it means that the instruments have sunk or that the staves have risen; therefore: the instrument must be set up at a more stable place.
- if the signs are always negative, the instrument has risen or the staves have sunk; therefore: the ground-plates must be set more carefully and at more stable places.

(18) **Check:** (5) = (10)

If not, inform the observer at the next set-up to equalize the sums.

(19) (2) - (6) if staff A is backward

(6) - (2) if staff A is forward

(20) (14) - (11) if staff A is backward

(11) - (14) if staff A is forward

- (21) **Check:** (20) + (17) = (19)
 (22) Mean of (19) and (20)

If there are any gross errors in the readings, or if the readings are out of tolerances, the booker informs the observer **after finishing all readings on that set-up** and not somewhere in between readings.

If the booker finds all readings to be correct after performing **all** of the above mentioned checks up to step (18) and if **all** data of that set-up are **recorded**, the booker gives the command "Finish move."

In the column **Remarks** of the field-book all the unusual happenings outside the normal procedure should be recorded, especially:

- condition of BMs
- steel ball, road nail or any other points used for turning point
- Sketch showing how a triple group and Trig-pillar was levelled.
- sudden change of weather
- problems in observations (e.g. shimmering)
- Beginning and end of a 14-days section etc.

9.7 Sum Foresight-Distances = sum Backsight-Distances

The subsequent positions of the instrument and the staves are already marked by the reconnaissance party, making the Backsight equal to the Foresight. However, differences may be discovered during observations by measuring the distances accurately with the instrument.

In such a case, the booker commands the observer to slightly alter the marked instrument position on the next set-up so as to equalize the sum of Backsight-Distances and the sum of Foresight-Distances.

Example:

Distance			
Back		Fore	
78	72	78	61
18	16	19	29
96	88	97	90

As the sum of Foresight distances is 1m longer than the sum of Backsight distances the booker commands the observer.

"Backsight 0.50m longer:"

Note:- Only **half of the difference** must be corrected to equalize the sums.
 As to avoid confusion, the correction necessary is to be given always in terms of the backsight.

The positions of the **last instrument set-up** between two BMs (PBM or TBM) is to be determined in such a way, that the sum of Backsight distances and the sum of Foresight distances between the two BMs become equal to within $\pm 0.020\text{m}$.

9.8 Moving to the next set-up

The instrument is protected against the sun with the instrument cover, while moving to the next set-up.

After reading the temperature at staff A, the staff supervisor sends the chainman with that staff to the next position and he himself goes to staff B. He revolves this

staff so that it faces exactly towards the new instrument position. He checks the staff's circular bubble. He now goes to the position of staff A and establishes it over the mark

- Note:-** 1. Take extreme care of that staff which keeps its position for the next set-up; i.e. staff in the Turning point.
2. If the line of levelling leads over vegetable mounds, this is to be dug out to set the ground plates and the tripod on gravel or sand, which are the best bases for stable set-ups.

In this set-up the same sequence of observation is repeated as described earlier.

Note:- for the relevant sequence of observation and booking when staff A is on Backsight or on Foresight see fig. 39.

Starting from a fixed point (PBM/TBM) the sequence of observation is repeated at every set-up until it reaches another fixed point (PBM/TBM). This is the process of levelling in the field.

9.9 Specification

	H.P./1st order	2nd order	3rd order
Level Instrument	Wild N ₃ or equivalent	wild N ₃ or NA ₂ with parallel plate micrometer or equivalent	Wild N ₂ or NA ₂ with parallel plate micrometer or equivalent
Staff	3m Invar staves with double graduation	3m Invar staves with double graduation	Invar staves or equivalent
Length of sight (max)	25m	30m	35m
Double levelling	Yes	Yes	Yes
Forward and Backward Levelling is carried out by	Different instrument and different observer	Different instrument and different observer	Different/same instrument and different/same observer
Observation time	Morning and Afternoon	Morning and Afternoon	Wholeday
Tolerance for Back and Fore Levelling (distance K in km)	$\pm 1.1 \sqrt{K}$ mm	$\pm 2.0 \sqrt{K}$ mm	$\pm 5.0 \sqrt{K}$ mm
Number of set-ups between BMs	even	even	even

10. ERRORS AND THEIR ELIMINATION

There are three different kinds of errors:

- (a) Gross errors
- (b) Systematic errors
- (c) Random errors.

10.1 Gross errors

These errors are caused by carelessness. They really are errors in the proper sense of the word and can be avoided by careful work. Above all, they are **discoverable** by repeating the measurements or by check-computations.

Examples:

- (a) Reading to the left scale of staff A: 87.746 cm
Reading to the right scale of staff A: 391.302 cm
The difference is 303.556 cm
The difference should be 301.550 cm

Here we can see that the gross error is introduced in one of the above two readings. To determine the gross error the readings are to be repeated.

- (b) A 223.65 m. long sub-section was levelled forward and backward. The difference of height between the two BMs was determined by team I as 81.72364 m. and by team II as 83.65738 m.
difference = 1.93374 m.
tolerance = 0.00019 m.
It is obvious, that either one or both of the differences of heights are grossly wrong.

Notes:- (1) Gross errors can occur, where ever observations are taken.

- (2) They are disclosed by **two independent observations** of the same values (staves with double graduations, Forward and Backward Levelling, etc.) and by careful execution of **computing checks**. Each computation is to be carried out by two different persons.

- (3) The procedure of the levelling work as described in this instruction book allows one to discover each and every gross error present in the field.

- (4) Therefore: **Gross errors should not exist in any datas sent from field to the headquarter.** This is one of the responsibilities of the Group Leader. He cannot delegate this responsibility to anybody else.

10.2 Systematic errors

A systematic error is one that occurs always with the **same sign**, and often with a similar magnitude in a number of consecutive or otherwise related observations. Systematic errors are caused by:

- (1) Poor measurement (e.g. staff not on stable ground)
- (2) The influence of the atmosphere (refraction)
- (3) Badly adjusted instruments.

This error is one of the most dangerous things in the process of levelling. Repetition of an observation does little or nothing to reduce the ill effect of **systematic errors**, which therefore **are a most undesirable feature** in any

observations. The influence of the systematic errors may be reduced to a certain extent by:

- (1) Establishing and **following** a correct measuring sequence.
- (2) Taking into account mathematically the previously determined extent of their influence.
- (3) Careful adjustment of the instruments.

10.2.1. Staff graduation error

This is the error of the graduation of the staff, caused by elongation or contraction of the invar tape due to change of its temperature and tension.

This variation in length of the staff **is to be checked.**

- (1) At the beginning and end of each field season.
- (2) **Immediately after the staff falls down.**

The calibration of the staves are carried out at the beginning and the end of a field season by the Double Image Comparator at head quarters.

In the field, whenever necessary the calibration is carried out by an Invar (nickel-steel) Testmeter (R. & A. Rost). This is a rod of 1110 mm length, 20 mm width and 10 mm thick. Near each end of it a fine line is engraved.

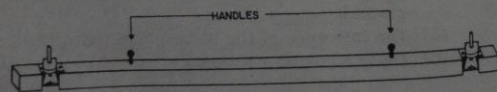


Fig. 40

On either side of both lines, which are $(1.0 \pm c)$ mm apart, (see **certificates** of each Testmeter for the exact correction c) 10 graduation lines at an interval of 0.1 mm are engraved, i.e. there is a two mm long scale at each end of the Testmeter. A microscope is fixed above each scale

which allows direct reading of 1/10 mm and an estimation up to 1/100 mm.

The calibration of the staves, i.e. the investigation of the deviation of their graduations from the international metre is to be performed as follows:

- (1) This test is to be carried out **in shadow.**
- (2) The staff (staves) which are to be tested are placed in shadow **at least half an hour before the test.**
- (3) The staff is to be laid horizontally, supported **at both ends and in the middle.**
- (4) The Testmeter fits exactly into the hollow portion of the staff, so that it can be placed directly on to the invar tape.
- (5) The engraved scale (B) on the right side of the Testmeter is placed over that line on the invar-tape of the staff, **which is numbered as 10** on the staff. The left hand graduation thus falls on the line of 110. (see fig. below)
- (6) Viewing through the microscopes, the readings are taken as shown in fig.

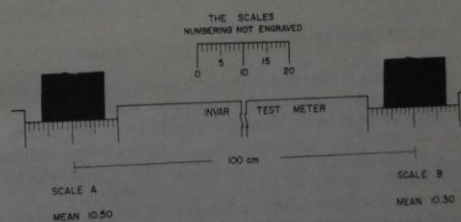


Fig. 41

- (7) The readings are taken on the left **and** on the right border of the line. **Without touching** the Testmeter, the readings are taken on both microscopes, i.e.

four readings at each position of the Testmeter. The booking is done in Lev. Form 1.10.

- (8) The Testmeter is now displaced by 10 cm, i.e. to the number 20 and 120 respectively. **Caution. While displacing the Testmeter, lift it slightly off the invar-tape in order not to rub off (with the time) its graduation.**
- (9) In this way **19 positions of the Testmeter are observed** the last being at the numbers 190 and 290 respectively.
- (10) In the similar way the graduation of the "big" numbers on the same staff are observed, beginning at the numbers 310 and 410 respectively and ending on 490 and 590 respectively.

Note:- Readings and bookings are to be performed in units of 1/10 mm, the distance between two consecutive lines of the graduation is 1/10 mm.

The temperature is to be recorded in the Lev. Form 1.10 immediately before each displacement of the Testmeter.

10.2.2. Zero error

This is an error if the zero point of the graduation of a staff is not exactly at the base of the staff.

This error is to be checked:- (1) Before beginning of each 14-days section.
(2) after any accident with the staff.

In levelling, the difference of height between two consecutive BMs are measured with the help of staves. The zero error is eliminated if it is started and end up by the same staff, but it has to be applied if a section is ended by a different staff.

Zero error = (Reading on staff A) - (Reading on staff B). For this purpose the staves are set up at the same BM and the readings are taken for both staves from the **same position of the instrument.**

The investigation of the zero error is to be performed as follows:

- (1) Establish the instrument at about 10 m from a PBM.
- (2) Staff A* at the PBM.
- (3) Reading a_1 on both scales of staff A and check **301.550.**
- (4) Staff B* at the same PBM.
- (5) Reading b_1 on both scales and check.
- (6) Again staff A at the PBM.
- (7) Reading a_2 on both scales and check.
- (8) If reading $a_2 = \text{reading } a_1 \pm 0.003 \text{ cm}$ (to ensure that the instrument has not moved)
Then zero error = reading a_1 minus reading b_1
 $= a_1 - b_1$.
If a_2 is not $a_1 \pm 0.003 \text{ cm}$, then repeat steps 1 to 8.
- (9) The booking is performed in Lev. Form 1.12.

* The letters are written on the staves by the factory.

10.2.3. Collimation error

This is the error caused by the line of collimation (line of sight) not being horizontal when the split bubble is centred, i.e. if the split bubble is out of adjustment.

The test of the horizontality of the line of collimation (line of sight) is carried out:

- (a) at the beginning of each 14-days section.
- (b) immediately after each accidental hard shock of the instrument.

- (c) Never in the line of levelling, but always after or before carrying out the actual levelling work.

The test is carried out as follows:

- (1) In more or less **flat terrain** a distance between 45 and 60 m long is selected and **divided accurately** with the aid of a tape into three **equal** sections of the length d . The points A, B, C and D are marked on the ground.

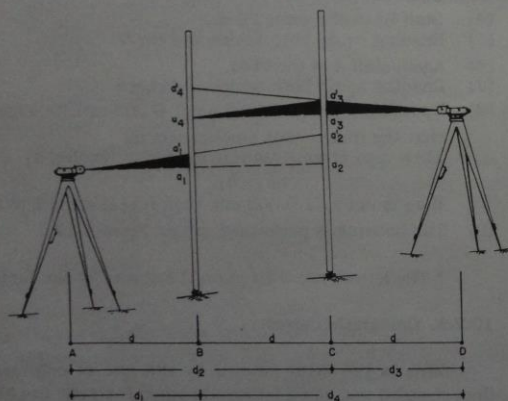


Fig. 42

- (2) The invar staves are set up on an iron ground plate at the intermediate points B and C.
- (3) The instrument is first set up at A (centred accurately above the mark) levelled up and the staff

readings a_1 and a_2 as well as the stadia readings for the distances are observed and booked in Lev. Form 1.13.

- (4) Set up the instrument at D and record the staff readings a_3 and a_4 .
- (5) **Check:** (distances from fieldbook)
 $2 d_1 = d_2 \pm 10 \text{ cm}$ and $2 d_3 = d_4 \pm 10 \text{ cm}$
 and $d_1 = d_3 \pm 10 \text{ cm}$ and $d_2 = d_4 \pm 10 \text{ cm}$
 If not, correct the distances accordingly and repeat all observations.
- (6) In fig. 42
 a_1, a_2, a_3, a_4 are the readings on the staves if the line of sight is absolutely horizontal.
 a'_1, a'_2, a'_3, a'_4 are the observed readings on the staves (here the line of sight is inclined).

If an imaginary line parallel to $(a'_1 - a'_2)$ is projected through a'_3 , it will cut the staff at B in the correct position a_4 , thus giving the required value for a true horizontal line of sight from D.

All shaded triangles are **congruent** and a'_1, a'_2, a'_3 and a'_4 is a parallelogram. Therefore

$$a_4 - a'_1 = a'_3 - a'_2$$

$$a_4 = a'_1 - a'_2 + a'_3$$

It is seen, that the correct value a_4 can be computed from the "wrong" observed values.

- (7) The computation of a_4 is carried out in Lev. Form 1.13, $(a_4 - a'_4)$ is the collimation error for the

back to zero and the level must be adjusted as described in 10.2.3.2.

- (9) The test of collimation error is to be carried out once again, to see whether the adjustment was done correctly or not.

Note:- For the execution of an adjustment of the collimation error, the **staff at B** and the **instrument at D** must keep **exactly the same position** as they had been during collimation error checking.

10.2.3.2 Adjustment by means of tubular level

If the value of **c dash (c')** is greater than **3mm** the adjustment is carried out as follows:

- (1) The staff at B and the instrument at D are still exactly in the same position as described in 10.2.3.
- (2) The micrometer reading is set to the computed millimetre value of a_4 .
- (3) The staff at position B is sighted (See Fig. 42).
- (4) The **tilting screw** is turned until the required staff centimeter graduation (of a_4) is straddled symmetrically and accurately by the wedge-shaped reticule lines.
- (5) In doing this, the ends of the split-bubble are no more in coincidence. The perfect coincidence of the split-bubble is now restored by means of the **vertical adjustment** of the level vial holder (see fig. 44).
- (6) The outer horizontal screw must be loosened slightly, so that the bubble ends can follow the adjustment movements of the vertical screws.
- (7) If the **left** bubble end is **low**, the lower adjustment screw is **loosened** slightly and the upper one is **tightened** by **equal** amounts as seen from the

objective, **both** are turned to the left (i.e. both always in the same sequence).

- (8) The adjustment is continued in this way, **step-by-step**, until the bubble ends are in coincidence. Turn the adjustment screws **always a minute amount only**.
- (9) The outer horizontal screw, originally loosened, is again tightened.
- (10) Check that the micrometer reading is still correct (a_4) when the bubble-ends are in coincidence.
- (11) Finally, the horizontality of the line of sight of the **adjusted instrument** is re-checked, to see whether the adjustment was performed correctly or not.

Note: (a) This adjustment is **not easy**:

- (b) All the turned screws should be tightened carefully and moderately, otherwise the adjustment will not last. They must not be tightened **too much**.
- (c) The last movement of any screw is always in its tightening sense.

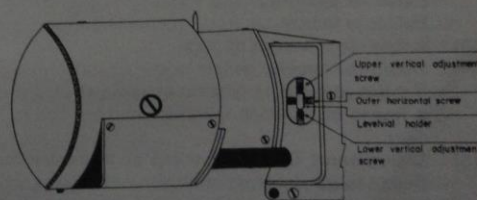


Fig. 44

The collimation error of the instrument thus determined can be applied in the observed difference of height or can be cancelled by keeping the Backsight and the Foresight distance equal.

10.2.4 Non-vertically of the instrument's vertical axis

The vertical axis of the instrument is set vertically by centering the instrument's circular bubble. If the circular bubble is not in correct adjustment, the vertical axis of the instrument can no more be set vertically. This will hamper speedy work and it may cause small errors during observations.

Therefore, the adjustment of the instrument's circular bubble is to be checked. The check is performed before the beginning of each 14-days section.

Check and adjustment

- (1) Set up the tripod, with instrument attached, on stable ground and protect it from the sunshine by an umbrella.
- (2) Using the three footscrews the circular bubble is centred **exactly** in the middle of the black setting circle. A **parallax-free** viewing of the circular bubble is to be made in the rotatable mirror.
- (3) Rotate the telescope by 200^g .
- (4) Watch the circular bubble in the rotatable mirror. If it has changed its position by **more than 1 mm**, it has to be adjusted otherwise nothing has to be done.
- (5) **Adjustment:** Clamp the telescope in its position. Do not touch the circular bubble with fingers.
- (6) **Exactly half** of the bubble's displacement from the centre is to be corrected by the footscrews.

Example: Note that the bubble must move along the straight line while turning the footscrews.

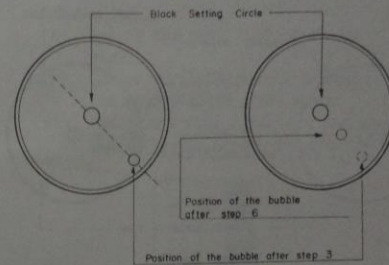


Fig. 45

- (7) The rest of the displacement, i.e. the **other half** of it, is to be corrected by the adjusting screws (at the bottom of the circular bubble), using the adjusting pin as follows.
- (8) **The first adjustment screw to be turned**, is the one that is the **nearest** to being in line with the middle of the bubble and the centre of the black setting circle. It is turned slowly, until the bubble reaches the centre of the circle or until the centre of the bubble reaches the nearest possible line from the centre of the circle to any of the other two adjusting screws.
- (9) If the bubble cannot be brought into the centre by step 8, then the next adjusting screw to be

turned is the one, that lies in the same line as the centre of the circle and the centre of the bubble.

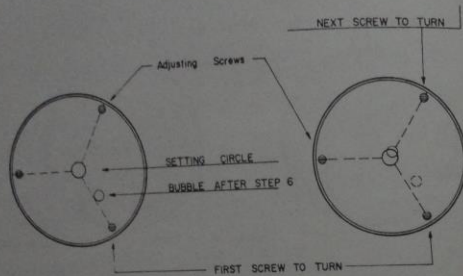


Fig. 46

Note: The screws must not be turned more than is necessary to complete the adjustment.

Check of the adjustment

Rotate the telescope always a small amount only and check after each such small movement the position of the circular bubble. It must now remain in the centre, whatever direction the telescope is pointing.

10.2.5 Non-verticity of the staff

The staff is set vertically by centering its circular bubble. If the bubble is not in correct adjustment, the staff cannot be set vertically. This means that each reading was taken too high on the staff.

Therefore, the adjustment of the staff's circular bubble is to be checked. The check is to be carried out

before the beginning of each 14-days section.

(a) Check and adjustment by the help of round box bubble

- (1) Set up the staff, well stable on a ground plate.
- (2) Centre the staff's circular bubble by turning the handles of the staff's struts.
- (3) Rotate the staff a full turn and direct it so that its bubble is placed in shadow.

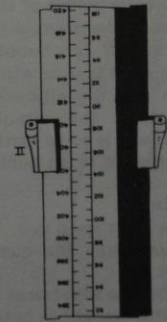


Fig. 47

- (4) Take a round box bubble and hold it firmly against a corner of the staff, approximately at the height of the staff's circular bubble. This being position I of the round box bubble. Be careful, **not to bend the staff** in doing so. Level the round box bubble by turning the handles of the struts.

- (5) **The same round box bubble** is now held firmly of that corner of the staff, which lies diametrically opposite to the former one. This being position II.
- (6) If the round box bubble is still levelled in its position II, go to step 8.
- (7) If the round box bubble is no longer levelled in position II, the half of the deflection is corrected by turning the handles of the struts. (Compare with 10.2.4. step 6)
- (8) **The circular bubble of the staff** shows now its true displacement of the bubble. The full deflection of the bubble from the black setting circle is to be adjusted by the help of struts and the adjusting screws on the bottom of the circular bubble. The adjustment is to be carried out only, if the displacement of the bubble is **more than 1 mm.**
Adjustment.
- (9) See 10.2.4. steps 6 to 9.
Do not touch the circular bubble with fingers.
Check of the adjustment
- (10) Hold the round box bubble in position I as mentioned in step 4. **Remember** now the position of its bubble.
- (11) Hold **the same** round box bubble in position II. If now the deflection is the same, but in exactly opposite direction as in step 10, the adjustment was carried out correctly, otherwise the whole procedure must be repeated.

(b) **By the help of plumb bob**

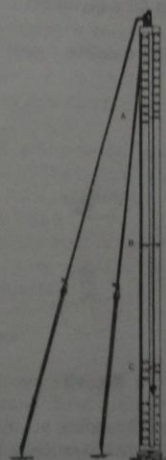


Fig. 48

This test is only possible when there is no wind or breeze in the field. In this test a plumb bob with a sufficiently long string is hung from the top of the staff, set vertically by the help of the circular bubble in such a way that the plumb bob hangs freely in the air (as shown in the figure) making the string and face of the staff parallel to each other. Check visually $AA' = BB' = CC'$.

If it is same i.e. plumb bob is parallel to the staff, the staff is vertical and the staff bubble is in

adjustment, otherwise the bubble is out of adjustment. This method is used only for checking the bubble but not for adjustment.

(c) By the help of Theodolite

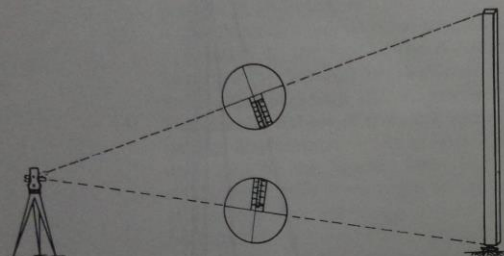


Fig. 49

- (1) Set the theodolite at about 15-20m, from the staff held vertically by the help of the circular bubble.
- (2) Bisect the base of the staff in such a way that the vertical cross hair coincides with the edge of the staff. Now check the verticality of the staff by slowly raising the telescope till it reaches the top of the staff.
- (3) At the top, if the vertical cross hair remains in the same position as it was in the bottom, the staff bubble is in adjustment in the direction perpendicular to the line of sight of the telescope.
- (4) If there is any displacement 'd' the bubble is out of adjustment.

Adjustment of the staff bubble

The displacement 'd' seen is eliminated by inclining the staff in the appropriate direction (perpendicular to the direction of the line of sight). By doing this if the displacement of the circular bubble is more than 1mm, adjust the bubble as described earlier. (see 10.2.5 step 7,8 and 9)

- (5) After adjustment, check this verticality again till the bubble is in adjustment.
- (6) Now turn the staff by 100° and repeat the steps 2 to 5.
- (7) When the staff is vertical in both directions then only will the bubble be in perfect adjustment.

10. 3 Random errors

They have different algebraic signs (+ or -). In a large number of observations there will be the same number of errors with a positive sign as there are errors with a negative sign. Also the magnitude of a random error is different for most observations. In contrast to gross errors and systematic errors, the random errors **cannot be avoided**. They are a result of the imperfections of the human senses and of the instruments. Even after adjustments the instrument will not be free of all errors.

10.3.1 Cross error

This error is defined as the difference between the direction of the tubular level and the direction of the line of sight of the telescope, i.e. the amount of horizontal non-parallelity between the two.

The test (and if necessary the adjustment) is to be carried out before the beginning of each 14-days section. **Before** the investigation of the cross error, the collimation error (10.2.3) and the instrument's circular bubble (10.2.4) have to be checked and if necessary adjusted.

Test

- (1) Set up a staff **at a PBM**.
- (2) Set up the instrument over the rigid type tripod 20m from the staff. The footscrews of the instrument are oriented so that any two footscrews are in a line perpendicular to the telescope when the telescope is directed towards the staff.
- (3) Centre the split-bubble, using the tilting screw. **Watch the bubble for a while**, to check whether it remains centred or not. If it does not, check instrument stability and repeat steps 2 and 3.
- (4) Book the following readings into Lev. Form 1.12. Stadia hairs and small numbers only (micrometer twice).
- (5) Turn the tilting screw a half turn. Set the split-bubble again in coincidence and read again, then book the small numbers once again (micrometer twice). The booker computes the mean m_1 of all readings of steps 4 and 5.
If v is greater than $\pm 0.005\text{cm}$, step 4 and 5 are to be repeated.
- (6) **The footscrew on the left of the eyepiece is now given a half turn.**
- (7) The micrometer reading is set to the computed millimetre value of m_1 .
- (8) Looking through the telescope, the **right footscrew** is now turned very slowly until the

centimetre graduation of m_1 is once again straddled symmetrically by the wedge-shaped reticule lines. Check again the micrometer setting.

- (9) Centre the split-bubble using the tilting screw.
- (10) Execute steps 4 and 5. The booker calculates the mean m_2 of these four readings.
- (11)

$$\text{Compute } e = \frac{m_1 - m_2 \left(\ln \frac{1}{100} \text{ mm} \right)}{\text{distance (in m)}}$$

- (12) If e is smaller than ± 1.5 the test is finished and no adjustment has to be carried out as the cross error in this case is considered as negligible.

Adjustment

- (13) The micrometer reading is set to the computed millimetre value of m_1 .
- (14) Using the tilting screw, the centimetre graduation of m_1 is straddled symmetrically by the wedge-shaped reticule lines.
- (15) The ends of the split-bubble are now to be brought in coincidence by turning the horizontal adjusting screws of the tubular level. The procedure is as follows (See Fig. 44).
- (16) One adjustment screw is **loosened first** slightly and the other is tightened by an **equal** amount (both being turned in the same direction).
- (17) The adjustment is continued in this way, step-by-step, until the bubble ends are in coincidence. The screws should be **tightened carefully and moderately**, otherwise the adjustment will not last.

- (18) Once the bubble ends are in coincidence, the whole test is to be repeated again (centre the circular bubble first) to see whether the adjustment is carried out successfully i.e. to see whether e is now smaller than ± 1.5 . If it is not, the adjustment is to be repeated.
- (19) When e is smaller than ± 1.5 the collimation error (10.2.3) is to be rechecked and, if necessary, be adjusted again.

11. STAFF AND THEIR DUTIES

11.1. Allocation of staff

Levelling Group

- 1. Group Leader
- 1 Team Leader
- 1 Computer
- 1 Kharidar
- 1 Driver
- 1 Orderly

Team I

- 1 Assistant Team Leader (Observer)
- 1 Booker
- 1 Staff Supervisor
- 1 Recce Supervisor

Team II

- 1 Assistant Team Leader (Observer)
- 1 Booker
- 1 Staff Supervisor
- 1 Recce Supervisor

The field work is carried out in two stages.

1. Reconnaissance and monumentation
2. Observation

For the first stage the Group Leader allocates the field personnel as per his own discretion. For the second stage the allocation is as follows:

Observation Sub-team	1	Assistant Team Leader (as observer)
	1	Booker
	1	Staff Supervisor
		Chainmen (as helper)
Marking Sub-team	1	Recce Supervisor
		Chainmen (as helper)

Since the observations are carried out by two teams (forward and backward direction) the above mentioned setting will be applicable for both teams.

11.2 Allocation of work and responsibilities

Group Leader : His direct superior is the head of the Levelling Division. He is responsible for:

1. The reconnaissance of Base Points and Permanent Bench Marks.
2. The organisation of the reconnaissance of all other Bench Marks and for all monumentations.
3. The quality of all data produced by his group.
4. The dispatching of all the field data (checked & completed) to headquarters.

He supervises and advises his Team Leader for the smooth running of daily work. He visits the field camp frequently in a month and submits the field report quarterly. He always emphasises on quality rather than on quantity.

Team Leader : His direct superior is the Group Leader. He is responsible for all the work carried out by his two teams and for their co-ordination. He issues the proper instructions to the respective observers to maintain the qualitative work. He signs daily the diaries of his observers. He rechecks the field book and other forms checked by the computer and sends them to the Group Leader. (He checks in

particular that there is no over-writing in the observed reading in any field books.) He is responsible for the smooth running of the field work.

Computer : His direct superior is Team Leader. He is entirely responsible for the technical side. He daily receives the field book from the booker after checking that it is neat and complete. He thoroughly checks the field book (see 9.6.1 reduction and checks) and entries the final ΔH and distance at the end of every sub-section. These results are then transferred to Lev. Form 1.16 for section checking. While checking the field books, any correction in the reduction column can be changed but the actual (original) observation may never be altered without any sound reason.

He is responsible for the checking whether the sub-section is inside the given tolerance or not. The field books of those sub-sections which are outside tolerance are to be thoroughly rechecked and then only they are to be listed in repetition list. After performing the required repetition, finally he checks the tolerance for all the sections. When all the measurements are satisfactory he then declares the end of the observation.

He should be critical in checking the field book and if any doubt arises he should consult his superior and should be strict for its quality control.

Observer :
(Assistant
Team Leader)

His direct superior is the Team Leader. He is responsible for all the work carried out by his team. The level instrument is his personal responsibility. He is responsible for the adjustment and safety of the instrument and for accurate observations. The instrument should be thoroughly checked, cleaned and protected when he is in-charge. While handing over to the next man, the instrument must be clean from dust and moisture.

The time of observation has been fixed according to the season by the consent of Group Leader and Team Leader and he is responsible for its execution. He allocates the work to be done in his team. If the observations are carried out without the stability of the instrument or staff and unequality of Back and Fore distances, such observations are to be cancelled and a new set is to be taken.

He maintains a daily diary containing all special events of technical interest that happened during the work.

Booker : He is responsible for all the observations (observed by the observer) to be booked neatly and correctly into the provided field book and to carry out the necessary computations and checks in the field book. Overwriting in the field book is strictly prohibited. If some reading after checking it twice, has to be corrected in the field, a completely new reading is taken cancelling the previous reading. The Back and Foresight distances are to be balanced at each set-up and at the last set-up of a sub-section it must be equalized. (See 10, Errors and their elimination). Each cancelled observation must have some sound reason testified in the remarks column. After completing the field book and testifying it with his and the observer's signature, he hands over to the computer daily without fail.

Staff Supervisor : He is responsible for both the staves and for their quick setting up at the reconnoitred places. If the reconnoitred place is not found to be stable he changes its position and later warns the recee supervisor about it.

He maintains the adjustment of the circular bubble of the staves. He checks the verticality of the

staves at each set-up before the first reading to any staff.

Proper handling of the staff is very important. If a staff has fallen down, the graduation and the adjustment of the circular bubble must be checked immediately thereafter.

Temperatures at each staff are recorded and conveyed to the booker for recording.

Recce

Supervisor : His job is to carry out the reconnaissance of the instrument and staff positions (see 5.4).

Proper reconnaissance (choosing the stable point for staff and instrument positions and equal Back and Foresight distance) is the vital part in achieving the qualitative work.

During reconnaissance he prepares a recce list which consists of: condition of Bench Mark, number of set-ups in the sub-section and any other remarks. After the team crosses, both recce supervisors interchange their list. Any damage of TBMs/PBMs are immediately informed to the staff supervisor.

12. MATERIAL AND MAINTENANCE

The materials required for levelling work have been categorised into two groups-

- (a) for reconnaissance and monumentation
- (b) for the observation.

The officer in-charge is responsible for checking all the materials required for his group before the work starts. Each item and everything are to be handled with special care. The Team Leader should check the materials before dismantling the camp. If there is any problem with the equipment or any damage or loss he should immediately report to his superior.

Maintenance:

Generally it is the duty of a surveyor to take utmost care of the instrument, which he is using. All the measurements depend on the instrument used, if the instrument is kept in good condition and is used with utmost care, one can achieve a very good result. One should never neglect the Survey instruments, these should be carefully used and maintenance should be carried out regularly.

This aspect of "maintenance or care of the instrument" is very vital for a surveyor, but most of them tend to neglect it. One is called a good and honest surveyor who loves one's instrument and keeps it always in a neat clean and working condition. It is not a hard job to clean and maintain the instrument in good condition, but one should feel that it is the part and parcel of his duty.

Instruments:

The following instruments at present are used for H.P/First Order and other Orders of Levelling.

- (a) Wild N₃ Precision level - for High Precision/First Order and Second order Levelling.
- (b) Wild NA₂ Automatic Level with parallel plate micrometer - This instrument may be used for Second and Third Order Levelling.
- (c) Wild N₂ with parallel plate micrometer - For Third Order Levelling.

Here the term maintenance means that the working instruments should be well looked after and therefore should be kept in working condition. If there are any repairs or adjustments to be made on the instruments, these should never be carried out in the field but should be sent to the mechanical workshop in the headquarters with a full report of the repairs or adjustments that are required.

For the field personnel when they receive the instruments, first of all they should go through the instruction manuals thoroughly and maintain the instruments accordingly. Daily cleaning and checking of all the instruments are of utmost importance.

Beside these, the observer should be aware of the following facts regarding the level instrument and Invar staves.

- (a) No liquid (i.e. water, kerosene, petrol etc.) should be used for cleaning.
- (b) Never touch any glass part (objective and eyepiece) with fingers.
- (c) While transportation the instrument should be carried on a soft pad or suspended.
- (d) If it is mounted on the tripod, always have somebody there.
- (e) Always store in a dust free, cool and dry place.
- (f) Always use the instrument cover (made of light cloth or plastic) while carrying the instrument during the observation.
- (g) Do not expose directly to sun, rain or frost, always use an umbrella to protect it.
- (h) Before starting the observation, the instrument should be taken out of the box at least half an hour earlier to adapt to the atmospheric temperature.
- (i) Invar staves and their numbering are to be cleaned regularly, but never scratch the invar band.
- (j) Regular greasing of struts should be carried out regularly.

13. FIELD PLOTTING AND COMPUTATION

Plotting

During reconnaissance and monumentation the alignment of the levelling traverse if it is not plotted in the map, will be plotted in a 1" = 1 mile map or the nearest available large scale map by careful map reading. The positions of all the Bench Marks established along the levelling line will be plotted after careful identification of all the local details in the map. The map may be made up to date by filling in the village name or town name and any other prominent feature as required. The position of the Bench Mark should be marked in ink by a circle and dot and numbered clearly and properly. When there are group of Bench Marks located so near that these cannot be shown separately in the map, in such a case a single dot and circle will signify all of them but the numberings are given just beside the dot. As shown below:

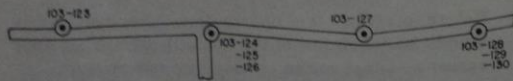


Fig. 50

Field Computation

The levelling procedures are adopted in such a way that the results are self checking and the errors are minimized. There is no chance of introducing any gross error in the levelling line, if the checking is performed seriously during the booking. The checks, which are to be performed during observation by the booker have already been described in chapter 9.

The error check forms (Lev. Form 1.10, 1.12 & 1.13) are completed and checked by the respective observer and booker and these are finally rechecked by the computer.

The field books duly completed and signed by the observer party are handed over to the computer. The level books thus received will have to be reduced at the conclusion of each days work and will not be left until the end of the section is reached.

Following are the list of checks and corrections to be applied:

- Thorough checking of the field book and mean difference of height.
- Application of distance correction (additive constant).
- Application of collimation and zero error.

After these checks and corrections the results are transferred to section checking form (Lev. Form 1.16). In this form, it is checked whether the levelling work carried out is within the allowable limit or not. If there are any sub-sections or sections, which are out of limits, these are sent for repetition until these are in the allowable limit. Once the check is performed the team moves to another section.

Keeping in mind that the field data sent to headquarters must be complete and error free, the completion of field data are rechecked by the Team Leader and the Group Leader. After the completion of the alignment, a technical report and statistical data (Lev. Form 1.29, 1.30) should be prepared and a check should be made to ensure that the work has been accomplished according to the assigned work programme.

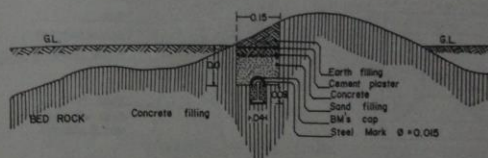
On satisfactory completion of a stretch or alignment, all level books should be sent by the quickest means to the officer in-charge for final checking and then forwarded to headquarters.

The list of technical data to be sent to headquarters
are listed in appendix L.

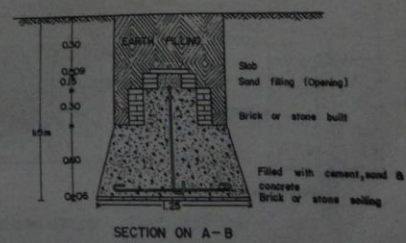
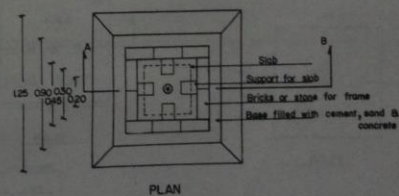
APPENDIXES

APPENDIX A/1

TYPE "A" BASE POINT ON BED ROCK

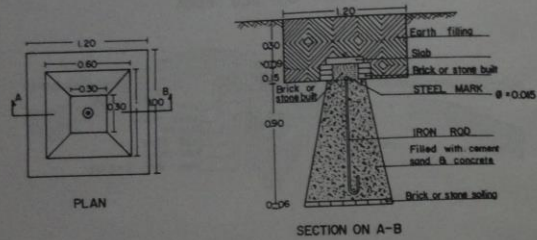


TYPE "B" BASE POINT ON CONCRETE FOUNDATION

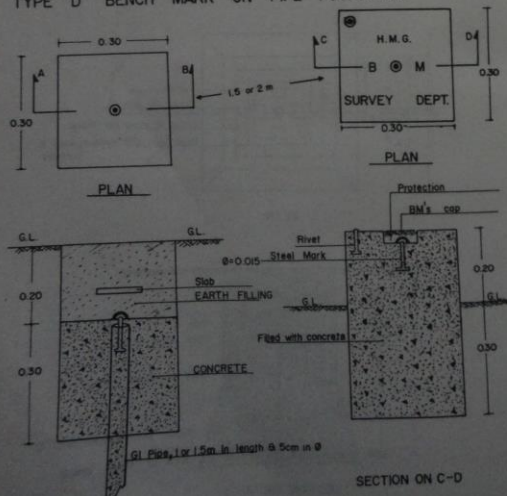


APPENDIX A/2

TYPE "C" BENCH MARK ON CONCRETE FOUNDATION

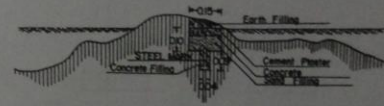


TYPE "D" BENCH MARK ON PIPE FOUNDATION

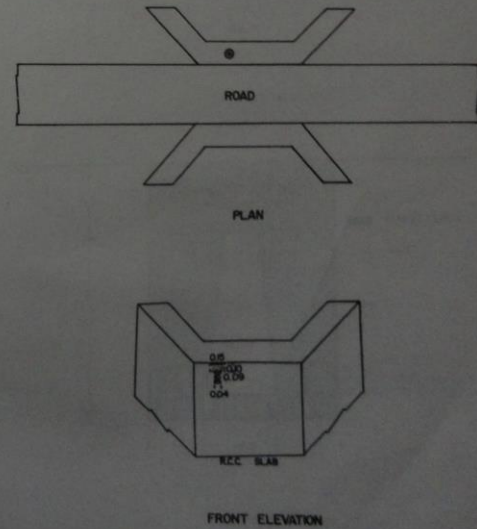


APPENDIX A/3

TYPE "E" PERMANENT BENCH MARK ON ROCK

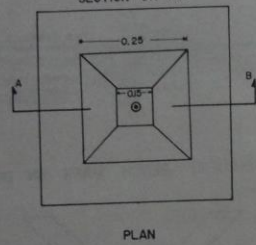


TYPE "F" PERMANENT BENCH MARK ON BRIDGE/CULVERT

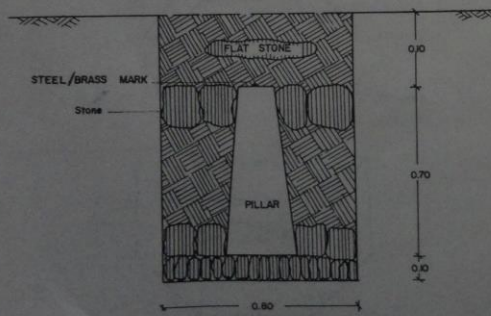


APPENDIX A/4

TYPE "G" PERMANENT BENCH MARK ON CONCRETE PILLAR
SECTION ON AB

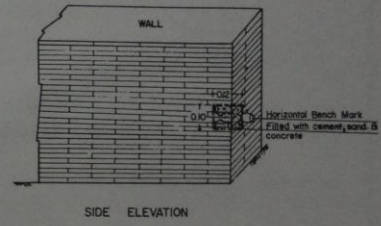


PLAN

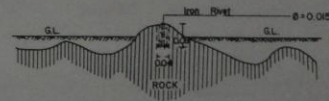


APPENDIX A/5

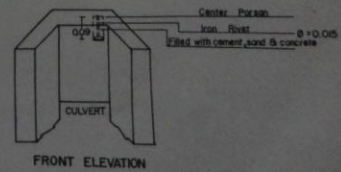
TYPE "H" HORIZONTAL BENCH MARK OR RIVET



SIDE ELEVATION



PLAN



FRONT ELEVATION

H.M.G. SURVEY DEPARTMENT DESCRIPTION OF BENCH MARKS

DISTRICT BHAKTAPUR PAGE NO. 14
 PANCHAYAT TUKUCHA GRID SHEET 102
 WARD NO. 7 ONE INCH SHEET 72 E/6

WRITTEN DESCRIPTION

105-027 - यो उचाई सम्राज विष्णु नगरकोटको रेडियो सण्ड टाइम सर्भिस हाउसको अगाडी जमिनको सतह भन्दा करिब 0.24 m तल स्थापना गरिएको छ।

105-028 - यो विष्णु नगरकोटको अवजरमेटोरी हाउसको चौदमा दक्षिण पश्चिम पट्टि जमिनको सतह भन्दा करिब 0.17 m तल स्थापना गरिएको छ।

105-029 - यो विष्णु नगरकोटको अवजरमेटोरी हाउसको उत्तर पश्चिम कुना पट्टि जमिनको सतह भन्दा करिब 0.5 m तल स्थापना गरिएको छ।

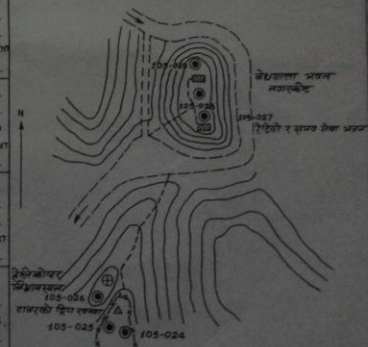
BM NO. 105-027
 TYPE: BRASS MARK / PILLAR ✓
 KIND OF LAND: UNCULTIVATED
 LANDOWNER: SURVEY DEPARTMENT
 SIGNATURE: _____

BM NO. 105-028
 TYPE: BRASS MARK / PILLAR ✓
 KIND OF LAND: UNCULTIVATED
 LANDOWNER: SURVEY DEPARTMENT
 SIGNATURE: _____

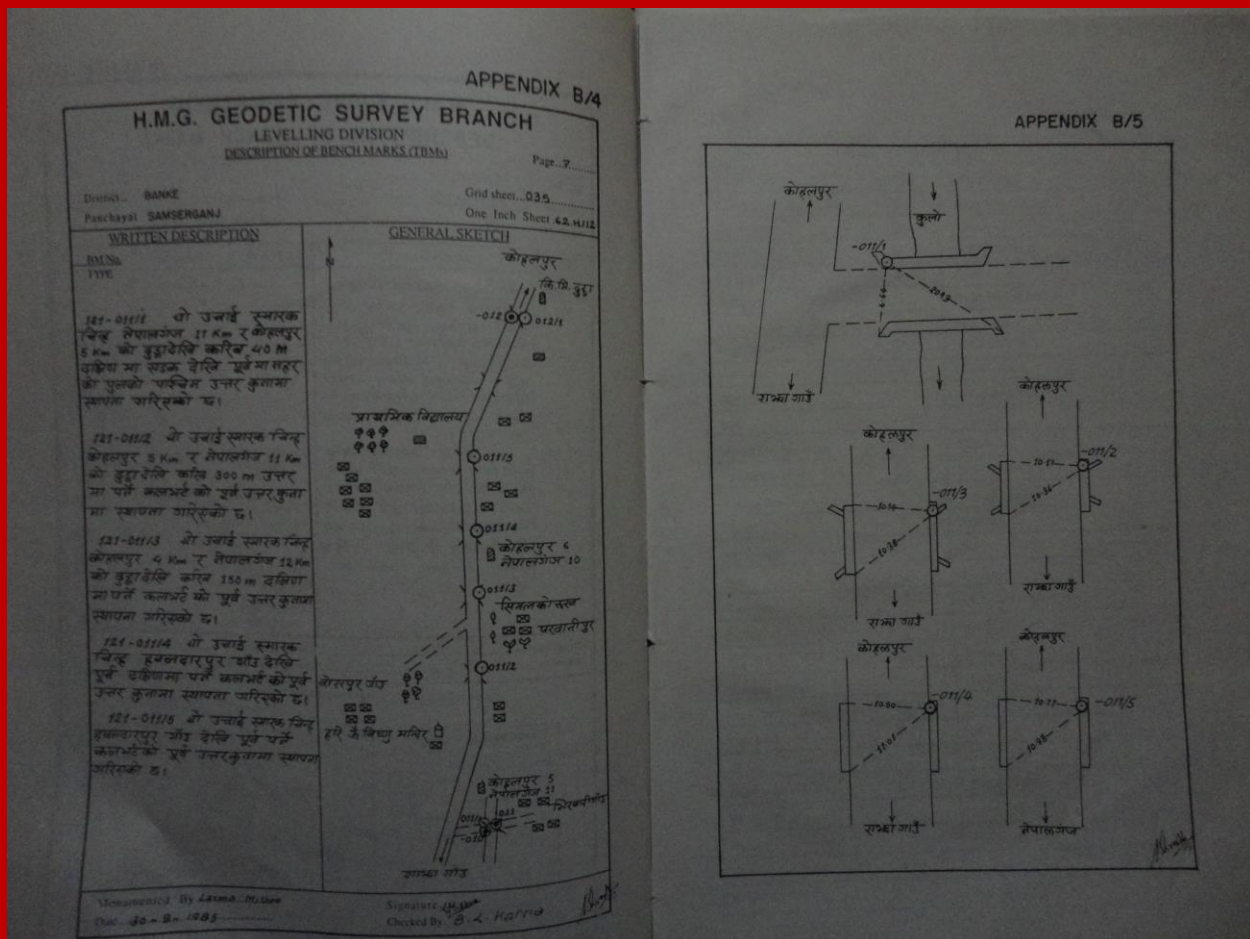
BM NO. 105-029
 TYPE: BRASS MARK / PILLAR ✓
 KIND OF LAND: UNCULTIVATED
 LANDOWNER: SURVEY DEPARTMENT
 SIGNATURE: _____

ACKNOWLEDGEMENT
 DATE 25 / 9 / 1976
 DONE BY: Chandra Bux Ulah
 SIGNATURE: _____

GENERAL SKETCH



327 FORM 125



Conventional sign			
Description	symbol	Description	symbol
Fundamental Bench Mark		Waterfall, rapids	
Datum Station		Railway	
Base Point		Traffic lights	
Permanent		Road metalled	
Temporary		Road unmetalled	
Trigonometrical station		Foot track	
Flag pole		Telephone line	
Mail box		Power line	
Road pointer		Water pipe line	
Gate		Chimney	
Milestone / Km stone		Tower	
Telephone booth		Hindu temple	
Post office	P.O.	Buddhist temple	
Pump		Statue	
Water tap		Dam	
Well		Forest	
Tube well		Dominant single tree	
Spring		Bamboo clump	
Customs office	C.O.	Banana plant	
Police station	P.S.	Garden	
Witness mark		Pond	
Brick built building		Lake	
Clay brick built building		Sandy land	
Canal		Sloping land	
Road over the river		Cliffs	
Bridge		Grassland	
Cuvert		Stone taps	
Railway over the river		Fence	
River, stream			

APPENDIX D

H.M.G SURVEY DEPARTMENT
LEVELLING DIVISION
RECORD OF BENCH MARKS

ALIGNMENT No. 120 FROM TYPE 'C' TO 120-017 Page No. 1

LOCATION	BENCH MARK		TOTAL		TYPE	DIST. Km/ Mile	PROTECTED BM No	REMARKS
	FROM	TO	PBM	TBM				
Chonoute	—		I		C	—	TYPE C	
“	Type C	118-056	I	—	F	0.531		
“	118-056	118-057	I	—	F	0.036		
“	118-057	118-058	I	—	F	0.096		
Birpur	118-058	120-001	I	6	F	3.257		
“	120-001	120-002	I	—	F	0.062		
“	120-002	120-003	I	5	G	2.118		
Jautharigaon	120-003	120-004	I	5	G	2.364		
“	120-004	120-005	I	—	G	0.060		
“	120-005	120-006	I	—	G	0.035		
Pulligaon	120-006	120-007	I	4	F	2.841		
Bahadurganj	120-007	120-008	I	4	G	1.964		
“	120-008	120-009	I	1	G	1.009		
“	120-009	120-010	I	—	G	0.130		
“	120-010	120-011	I	—	G	0.056		
Duyabagat	120-011	120-012	I	3	F	1.850		
Gharimpa Khola	120-012	120-013	I	6	F	2.724		
“	120-013	120-014	I	—	F	0.031		
Khusha Nagar	120-014	Type C	I	6	C	2.530	TYPE C	
“	Type C	120-015	I	—	F	0.309		
“	120-015	120-016	I	1	F	0.119		
“	120-016	120-017	I	—	F	0.098		
TOTAL			22	40		22.22		

Law Firm 128

APPENDIX E/I

H. M. G. SURVEY DEPARTMENT
LEVELLING OF HIGH PRECISION

[illegible]

GEODETIC SURVEY BRANCH
LEVELLING DIVISION

[illegible]

H.M.G. SURVEY DEPARTMENT
LEVELLING OF HIGH PRECISION
CHECKING OF SECTIONS

GROUP NO 2	FROM PBM	TO PBM
CAMP <i>Bariapa</i>	PAGE NO FROM	TO
TEAM I, FORWARD	TEAM II, BACKWARD	
OBSERVER <i>Rishi Ram Shrestha</i>	OBSERVER <i>Dinesh Prasad Karna</i>	
DATE FROM <i>30 / 12 / 1982</i> TO <i>1 / 1 / 1983</i>	FROM <i>22 / 12 / 1982</i> TO <i>24 / 12 / 1982</i>	

[illegible]

H.M.G. SURVEY DEPARTMENT

TOLERANCE TABLE

HIGH PRECISION/FIRST ORDER

TOLERANCE: $= \sqrt{12.1 D}$ 1/100 mm. D = Distance in metres

D TOL.	D TOL.	D TOL.	D TOL.	D TOL.	D TOL.	D TOL.	D TOL.	D TOL.	D TOL.		
3	82	057	264	057	549	082	937	107	4129	2054	2723
5	007	87	032	273	563	082	955	151	132	2050	2753
5	008	87	033	273	058	083	973	108	1473	2076	2783
6	009	93	034	293	059	084	991	109	154	2103	2814
7	010	98	035	306	060	084	1009	110	1517	2129	2844
9	011	104	036	303	061	086	1029	111	1537	2156	2875
11	012	110	037	313	062	087	1046	112	1560	2182	2905
13	013	116	038	323	063	088	1065	113	1585	2209	2937
15	014	123	039	333	064	089	1083	114	1608	2236	2969
17	015	129	040	344	065	090	1103	115	1631	2264	2999
20	016	136	041	355	066	091	1122	116	1655	2291	3031
23	017	142	042	365	067	092	1141	117	1679	2318	3063
25	018	148	043	377	068	093	1161	118	1702	2348	3094
28	019	154	044	388	069	094	1181	119	1726	2374	3126
31	020	161	045	399	070	095	1200	120	1750	2405	3159
35	021	171	046	410	071	096	1220	121	1774	2431	3191
38	022	179	047	423	072	097	1240	122	1798	2459	3224
42	023	186	048	434	073	098	1261	123	1823	2488	3256
46	024	194	049	446	074	099	1281	124	1847	2517	3289
50	025	203	050	459	075	100	1302	125	1872	2545	3321
54	026	211	051	471	076	101	1323	126	1897	2575	3352
58	027	219	052	484	077	102	1343	127	1922	2604	3383
63	028	233	053	509	078	103	1365	128	1947	2633	3424
67	029	237	054	522	079	104	1386	129	1973	2663	3454
72	030	245	055	536	080	105	1407	130	1998	2693	3490
77	031	255	056	556	081	106	1429	131	2024	2723	3524
82	032	264	057	549	082	107	1450	132	2050	2753	3559

APPENDIX H/2

H.M.G. SURVEY DEPARTMENT
LEVELLING OF HIGH PRECISION

INSTRUMENT No. 26444 DATE 21.8.1956 WEATHER Clear PAGE No. 5
OBSERVER Nil Govinda Maharaja BOOKER Bar. Nanyang Chaudhary
Determined value of α & β are valid for 14 days session from Bench Mark 111-112 to B.P. 112-3

CROSS ERROR (a)

STADIA	Reading	CENTRE HAIR	Mean	STADIA	Reading	CENTRE HAIR	Mean
U	132.132	132.132	132.132	51			
L	132.132	132.132	132.132	52			
STADIA	Reading	CENTRE HAIR	Mean	51-52			
132.132	132.132	132.132	132.132	Mean in M1			
132.132	132.132	132.132	132.132	53			
132.132	132.132	132.132	132.132	54			
132.132	132.132	132.132	132.132	53-54			
132.132	132.132	132.132	132.132	Mean in M2			
132.132	132.132	132.132	132.132	M1-M2			
132.132	132.132	132.132	132.132	D			

(M1-M2) = 0.00 mm/100

ZERO ERROR (x)

STADIA	Reading	CENTRE HAIR	Mean	STADIA	Reading	CENTRE HAIR	Mean
U	132.132	132.132	132.132	51			
L	132.132	132.132	132.132	52			
STADIA	Reading	CENTRE HAIR	Mean	51-52			
132.132	132.132	132.132	132.132	Mean in M1			
132.132	132.132	132.132	132.132	53			
132.132	132.132	132.132	132.132	54			
132.132	132.132	132.132	132.132	53-54			
132.132	132.132	132.132	132.132	Mean in M2			
132.132	132.132	132.132	132.132	M1-M2			
132.132	132.132	132.132	132.132	D			

STADIA: A, No. 26444
B, No. 26444

NOTE: 1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

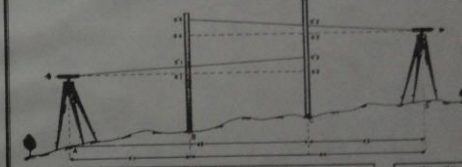
1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

APPENDIX H/3

H.M.G. SURVEY DEPARTMENT
LEVELLING OF HIGH PRECISION

INSTRUMENT No. 36666 DATE 20.1.56 WEATHER Clear PAGE No. 1
OBSERVER Nil Govinda Maharaja BOOKER Gunj Bahadur Basnet
Determined value of α is valid for 14 days session from Bench Mark 111-112 to B.P. 112-3

COLLIMATION ERROR & ITS CORRECTION (a)



STADIA	Reading	CENTRE HAIR	Mean	STADIA	Reading	CENTRE HAIR	Mean
U	132.132	132.132	132.132	51			
L	132.132	132.132	132.132	52			
STADIA	Reading	CENTRE HAIR	Mean	51-52			
132.132	132.132	132.132	132.132	Mean in M1			
132.132	132.132	132.132	132.132	53			
132.132	132.132	132.132	132.132	54			
132.132	132.132	132.132	132.132	53-54			
132.132	132.132	132.132	132.132	Mean in M2			
132.132	132.132	132.132	132.132	M1-M2			
132.132	132.132	132.132	132.132	D			

NOTE: 1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

1. Staff on P.M.
2. Distance from P.M.
3. Sequence: Staff A—B—A

H.M.G. GEODETIC SURVEY BRANCH

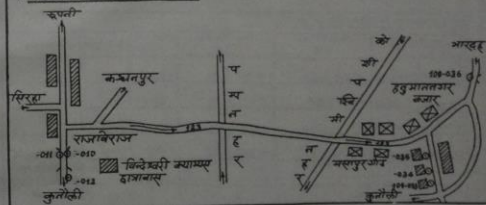
LEVELLING DIVISION

TECHNICAL REPORT

Page: 1

1. ALIGNMENT NO. 723, FROM 109-072 TO 109-036
 TOTAL LENGTH FORWARD, 15.61... KM DATE FROM 7.13.1988 TO 21.3.1988
 TOTAL LENGTH BACKWARD, 15.61... KM DATE FROM 6.13.1988 TO 21.3.1988

2. SKETCH OF THE ALIGNMENT



3. STARTING JUNCTION POINT

(i) Few words about its location, condition

109-010, 109-011 & 109-012 :- ये वैचार् स्मारक विन्ध्यपुर राजविराज-कुपली सड़क से जंदा महेन्द्र विन्ध्यपुर जयपुर को राजविराज मजिद को पुल को बनाये पड़े, सोहि पुल को सोहि पट्टि माथको जंदा र बायो पट्टि विन्ध्यपुर को पुल को जंदा कायम दूर। सहे वैचार् स्मारक विन्ध्यपुर रायो अवस्था मा दूर।

(ii) Comparison of (New-old) results

Bench Mark	New Result		Old Result	Difference (N-O)	Tolerance	Remarks
	From	To				
109-072	109-071	+ 50.592	+ 50.606	-12	45	700-072
109-071	109-070	+ 9.460	+ 9.549	-89	26	700-071
						700-070
						700-069
						700-068
						700-067
						700-066
						700-065
						700-064
						700-063
						700-062
						700-061
						700-060
						700-059
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						700-051
						700-050
						700-049
						700-048
						700-047
						700-046
						700-045
						700-044
						700-043
						700-042
						700-041
						700-040
						700-039
						700-038
						700-037
						700-036

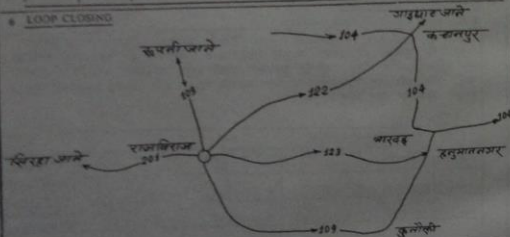
4. LEVELLING LINE

(Few words about the nature of the terrain, weather condition, overall condition of BM, Line of dissection BM and about observation)

ये लेवल लाइन को प्राकृतिक स्वरूप समझल दू।
 उरी लाइन जोर चले पक्की सड़क द। सर्वेक्षण समय मा मौसम आरि र सड़क द। मुझे स्मारक विन्ध्यपुर रायो अवस्था मा दूर। बल लेवल लाइन मा सर्वेक्षण कार्य बिहान 6:30 देखि 6:30 सम्म र मेसुमा 2:30 देखि 2:30 सम्म गरिएको द।

502-033, 502-034, 502-035 व. 502-036 :— श्री उमाई स्मारक विद्यालय
कुलाला-भारद्वय परचालन कमीती वार भारद्वय जोरा कमार हनुमाननगर
भारद्वय अंगण शालाकाची अजादी, लोकमान्य जयकाई घर अजादी, विठ्ठल
नगर की घर अजादी व हनुमाननगर कमार वार भारद्वय जोरा अजादीया
पाठ्य लेखिका कळामरीया कायम द्या. एके उमाई स्मारक विद्यालय राणी
अकबरा घर द्या.

Inch. Start		New Retain	Old Retain	Undersize	Tolerance	Remarks
From	To	ΔH	ΔH	(#-0)		
100-023	100-026	+ 72-358	+72-082	-126	31	
100-026	100-035	+72-238	+72-230	-692	29	
100-035	100-036	+75-608	+74-092	+692	183	100-033
100-038	100-036	+76-226	+726-202	+74	109	100-029-030 values are previous results



BRIEF REPORT ABOUT THE TECHNICAL MANPOWER INVOLVED			
	समन्वयन	मारी	सुंदर नं १
	मारी	अधिकृत	जमशेदा असाद पठाण
दोही प्रमुख	तामना बनार विस्	कमरुद	उज महदुर वलोन
डायरेक्टर	अमोल कौशल माथार	अर्धक	उज नं १ विस्
	संयोजक	शोकी नं १	संयोजक शोकी नं २
सुपरवेजर	वेनमनमन चौधरी	संयोजक	शोकी नं ३
	कमरुद	नकर	विरेड काम कर्मी
सुपरवेजर	कौशल विस्	सहाय सुपर अइडर	कमरुद चौधरी
दोही	सुपर अइडर	नं १	नं १
	सुपर अइडर	नं १	नं १

STATISTICAL DATA Please Fill in the following information: Group / Term No. 1 Camp No. 12345

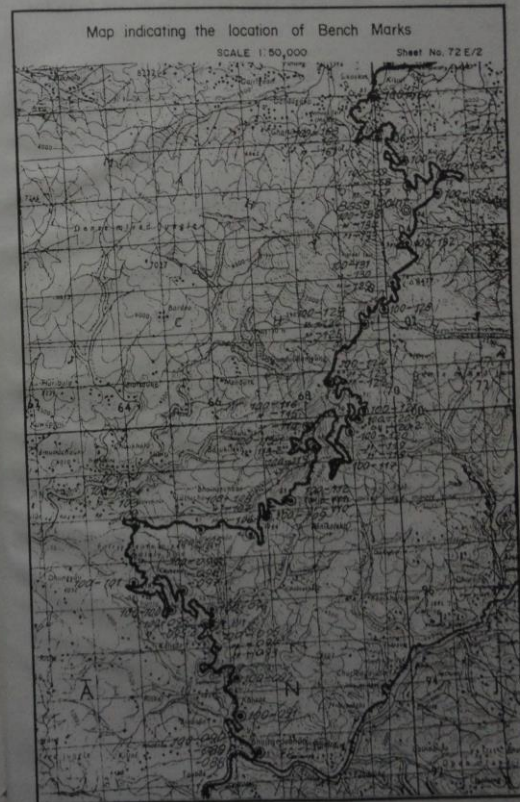
Prepared By: G. B. Biss... Date: 28/7/1968
Checked By: N. B. Bist...

App. Form 1.29

H.M.G SURVEY DEPARTMENT
LEVELLING DIVISION
STATISTICAL DATA

ALIGNMENT NO 123				DATE FROM 7/3/1988		20/3/1988		H.M. G SURVEY DEPARTMENT LEVELLING DIVISION STATISTICAL DATA		Page No. 1	
BM NO	TYPE	NO OF SECTION	INSTR- PERIOD (hr-min)	REPETITION DIST (km)	PERIOD (hr-min)	TOTAL DISTANCE (Kms)	DIFF- RANGE (1000ft) 1/200mm	ΣD 1/200mm	REMARKS $\sigma^2 = \frac{\sum D^2}{200 \times \frac{1}{2} \frac{1}{200}}$	r = NO OF SECTION D = DISCREPANCY BETWEEN FIB L = DISTANCE	
109-002	F	1	0 18			0 18	-34	-34	9600.00		
109-000	F	2	0 04			0 06	-35	-35	1225.00		
109-001	F	3	3 30			3 36	-18	-53	109.72		
103-001	F	4	1 55			1 55	+82	+82	2626.08		
103-002	F	5	1 55			1 55	+82	+82	2626.08		
103-003	F	6	0 16		0 14	0 30	-	-	216.67		
103-004	F	7	3 00		0 50	2 52	-150	-154	5648.04		
210-030	F	7	3 00		0 72	3 52	+7	+7	12725.00		
210-031	F	8	0 04			0 04			167.56		
210-032	F	9	0 11			0 11	-108	-108	2356.64		
103-005	F	10	3 03		0 58	2 56	+140	+95	17668.42		
103-006	F	11	3 11			3 11	-24	-119	9600.00		
103-007	F	12	0 08			0 08			167.56		
103-008	F	13	0 07			0 07			167.56		
103-009	F	14	2 03		0 33	2 36	-70	-108	2626.08		
103-010	F	15	0 04			0 04			167.56		
103-011	F	16	0 08			0 08			167.56		
103-012	F	17	0 04			0 04			167.56		
103-013	F	18	0 04			0 04			167.56		
103-014	F	19	2 31			2 31	+75	+137	3348.21		
103-015	F	20	18 43		1 79	2 02			3765.91		

1000



APPENDIX L

List of technical datas to be send to headquarters from the field

1. Description cards of PBMs with a duplicate copy
(Lev. Form 1.20)
2. Description cards of TBMs with a duplicate copy
(Lev. Form 1.22)
3. Observation books of Forward and Backward teams
(Lev. Form 1.11)
4. Checking of section with a duplicate copy
(Lev. Form 1.16)
5. Record of Bench Marks. (Lev. Form 1.24)
6. Error checking forms:
 (a) Collimation error. (Lev. Form 1.13)
 (b) Cross and zero error. (Lev. Form 1.12)
 (c) Staff standardisation. (Lev. Form 1.10)
7. Work programme.
8. Technical Report. (Lev. Form 1.29)
9. Statistical data. (Lev. Form 1.30)
10. Result of loop closing with diagram.
11. Map indicating the location of Bench Marks.